Supplementary material: eV-Scale Sterile Neutrinos Versus BBN and CMB $N_{\rm eff}$ Limits: Short-Baseline Anomaly Viability

HEP-Genesis-Agent session record
Project: ramp-research-sbl-neff

2026-11-07

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `0ffa6d3b`
- Started: 2026-11-07 17:16:16; report generated: 2026-11-07 18:22:52
- Project: ramp-research-sbl-neff (`/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff`)
- Compute execution site(s) selected in the session: local
- 2 user turn(s), 251 tool call(s) (6 returned errors), 4 figure(s), 11 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`muse-spark-bf16`	`https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1`
report	`muse-spark-bf16`	`https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1`
coder	`muse-spark-bf16`	`https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`

Server	Transport	Location
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`
paper-search	stdio	`python -m hep_genesis.paper_search`
web-search	stdio	`uvx duckduckgo-mcp-server`
cosmic-emulators	http	`https://cosmic.77-42-88-84.sslip.io/mcp`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`hep-research-aid list_agent_files`	`{}`	`[]`
2	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - papers - projects - reports - research-results - runs - scripts - ...`
3	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
4	`hep-research-aid list_datasets`	`{}`	`{"count": 0, "datasets": []}`
5	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`
6	`serve-sbbn.sh list_sterile_nu_skills`	`{}`	`{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...`

#	Tool	Arguments	Outcome
7	local web_search	{"query": "sterile neutrino short-baseline anomaly status 2024 2025 review"}	Jun 10, 2024 · Long-standing anomalous experimental results from short-baseline neutrino experiments have persisted for decades. These results, when interpreted with one or more light sterile neutrino...
8	local web_search	{"query": "eV sterile neutrino BBN N_eff limits CMB Planck 2024"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=eV%20sterile%20neutrino%20BBN%20N_eff%20limits%20CMB...
9	local web_search	{"query": "LSND MiniBooNE MicroBooNE sterile neutrino 2024 2025 arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=LSND%20MiniBooNE%20MicroBooNE%20ster
10	local web_search	{"query": "reactor antineutrino anomaly sterile neutrino 2024 2025 BEST gallium"}	Dec 14, 2024 · Reactor neutrinos are produced by the decay of fission products. The Reactor Antineutrino Anomaly (RAA) [1, 2] shows a 6.5% deficit in measured flux with respect to prediction at short ...
11	local web_search	{"query": "arXiv sterile neutrino global fit 2024 2025 short baseline"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%20sterile%20neutrino%20global%

#	Tool	Arguments	Outcome
12	local web_search	{"query": "arXiv MicroBooNE sterile neutrino search 2024 2025"}	Dec 8, 2025 · Here, we use data obtained from the MicroBooNE liquid-argon time projection chamber in two accelerator neutrino beams to exclude the single light sterile neutrino interpretation of the L...
13	local web_search	{"query": "arXiv BBN N_eff sterile neutrino cosmological constraints 2024"}	Jan 18, 2025 · The question of whether the massive sterile neutrinos exist remains a crucial unresolved issue in both particle physics and cosmology. We explore the cosmological constraints on the mas...
14	local web_search	{"query": "arXiv BEST gallium anomaly sterile neutrino 2024"}	Apr 23, 2025 · arXiv: 2504.16590 (hep-ph) [Submitted on 23 Apr 2025 (v1), last revised 29 Apr 2025 (this version, v2)] Title: Testing the Gallium anomaly using Electron-Neutrino Scattering Authors: Emi...
15	local web_search	{"query": "arXiv reactor antineutrino anomaly STEREO PROSPECT 2024 sterile"}	We describe an analysis of the full set of data generated by STEREO and an accurate prediction of reactor antineutrinos. The measured antineutrino energy spectrum suggests that the anomalies originate...
16	local web_search	{"query": "\"arXiv:2405\" sterile neutrino MicroBooNE exclusion 2024"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=%22arXiv%3A2405%22%20sterile%20neutr

#	Tool	Arguments	Outcome
17	local web_search	{"query": "\"arXiv:2412\" STEREO reactor sterile neutrino final results"}	Dec 26, 2024 · In this paper, we present an algorithm to generate the collider events of the GeV-scale oscillating sterile neutrinos with the ready-made event generation tools in the case that the cro...
18	local web_search	{"query": "\"arXiv:2501\" cosmological constraints sterile neutrino DESI Planck N_eff"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=%22arXiv%3A2501%22%20cosmological%20c
19	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
20	hep-research-aid get_dispatch	{}	Dispatch: local execution.
21	hep-research-aid list_agent_files	{}	[]
22	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
23	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
24	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1 ...
25	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [0.1, 0.5, 1.0, 1.14, 2.0, 5.0, 10.0], "sin2_2thetas": [0.0001, 0.001, 0.01, 0.05, 0.1, 0.42, 1.0], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/r...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/sterile_nu_grid_DW_1ddeb2.csv"], "message": "Predicted 49 points [...
26	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
27	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
28	local web_search	{"query": "MicroBooNE LArTPC 2024 2025 nu_e appearance disappearance exclusion LSND MiniBooNE 95% CL"}	Our new 2025 result on the e+e- Dark Sector Low-Energy-Excess search is available here. Our new 2025 results on the single-photon searches are available here. MicroBooNE is a large 170-ton liquid-argo...

#	Tool	Arguments	Outcome
29	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 0.4}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 0.4\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1 \u2...
30	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 0.001}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 0.001\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 0.39...
31	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.14, "sin2_2theta": 0.42}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1 ...
32	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 7e-11\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff ...
33	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 0.09}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 0.09\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1 \u...

#	Tool	Arguments	Outcome
34	local web_search	{"query": "STEREO final 119-day Dec 2024 PROSPECT STEREO combined 2024 2025 reactor antineutrino anomaly exclusion 99.9%"}	Oct 14, 2025 · STEREO's 119 reactor-on-day final dataset shows no L/E-dependent oscillation pattern in 2.5-7 m baselines and excludes the canonical eV-sterile parameter region of the reactor antineutr...
35	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [0.5, 1.0, 3.0], "sin2_2thetas": [0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.42, 1.0], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-res...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/sterile_nu_grid_DW_b2a7f6.csv"], "message": "Predicted 27 points [...
36	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 0.1, "m_max_eV": 10.0, "theta_min": 0.0001, "theta_max": 1.0, "n_points": 60, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/scan_DW_9bdb63.csv", "/Users/nramachandra/hep-genesis-testing/proje...
37	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 7e-11\n\nE_\u03b3 = 3.55 keV\nDecay...
38	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...

#	Tool	Arguments	Outcome
39	local web_search	{"query": "BEST gallium anomaly SAGE GALLEX 2025 2026 reanalysis 5 sigma sterile neutrino"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=BEST%20gallium%20anomaly%20SAGE%20GA
40	local web_search	{"query": "global 3+1 sterile neutrino fit 2024 2025 tension reactor gallium Planck ACT DESI N_eff 2.99"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=global%203%2B1%20sterile%20neutrino%
41	local web_search	{"query": "arXiv: 2504.16590 Testing Gallium anomaly Electron-Neutrino Scattering"}	Apr 23, 2025 · arXiv: 2504.16590 (hep-ph) [Submitted on 23 Apr 2025 (v1), last revised 29 Apr 2025 (this version, v2)] Title: Testing the Gallium anomaly using Electron-Neutrino Scattering Authors: Emi...
42	board post_finding	{"claim": "Canonical SBL best-fit sterile neutrino (m_s=1.14 eV, sin^2 2theta=0.42, Delta m^2=1.3 eV^2) fully thermalizes to Delta N_eff=1.0 (N_eff total 4.044) and is excluded at 6.2 sigma vs Planck+ACT+DESI N_eff=2.99±...	Recorded finding f-001.
43	board post_finding	{"claim": "Grid scan m_s=0.1-10 eV, sin^2 2theta=1e-4 to 1 (49 points, DW mechanism) shows near-complete thermalization Delta N_eff=1.0 for all sin^2 2theta >=0.01 at any eV mass, with 37/49 points BBN-excluded at 95% an...	Recorded finding f-002.

#	Tool	Arguments	Outcome
44	board post_finding	{"claim": "Anchor points quantify thermalization threshold: BEST-preferred m_s=1 eV sin^2 2theta=0.4 gives Delta N_eff=1.0 N_ eff=4.044 Omega_s h^2=22.0 excluded (Delta chi^2=23.1) ; reactor-like sin^2 2theta= 0.09 same N_e...	Recorded finding f-003.
45	board post_finding	{"claim": "BBN observables for fully thermalized sterile (N_eff=4.044) are Yp=0.2596 (3.7 sigma above observed 0.2449±0.004) and D/H=2.777e-5 (9.2 sigma above 2.547±0.025e-5), combined Delta chi^2=23.1 vs SM; tool validi...	Recorded finding f-004.
46	board post_finding	{"claim": "Thermalization threshold in standard Dodelson-Widrow cosmology: for eV-scale sterile neutrinos (m_s=0.5,1,3 eV), mixing sin^2 2theta >=0.003 yields Delta N_eff=1.0 (full thermalization), while sin^2 2theta=0.0...	Recorded finding f-005.
47	board post_finding	{"claim": "BBN/CMB exclusion plane scan (0.1-10 eV, sin^2 2theta 1e-4 to 1, 3600 points DW): 2597/3600 points BBN-excluded (Delta chi2>3.84, 95% CL) and 2362/ 3600 overclose (Omega_s h2> 0.12); maximum BBN-allowed mixing i...	Recorded finding f-006.
48	board post_finding	{"claim": "Canonical BEST SBL best-fit (m_s=1.14 eV, sin^2 2theta=0.42, Delta m2=1.3 eV2) predicts Delta N_eff=1.0 (N_eff=4.044) , Omega_s h2=29.3 (244x observed DM), BBN Yp=0.2596 and D/H=2.78e-5 with Delta chi2=23.1 vs ...	Recorded finding f-007.

#	Tool	Arguments	Outcome
49	board post_finding	{"claim": "keV sterile neutrino X-ray line contrast: m_s=7.1 keV, sin^ 2 2theta=7e-11 yields E_ gamma=3.55 keV, lifetime 5.8e27 s (1.8e20 yr), excluded by NuSTAR M31 and XMM-Newton blank sky; eV SBL mixing (0.1-0.5) is 9-1...	Recorded finding f-008.
50	serve-sbbn.sh scan_sterile_nu_ parameters	{"m_min_eV": 0.1, "m_ max_eV": 10.0, "theta_ min": 0.0001, "theta_ max": 1.0, "n_points": 60, "mechanism": "DW", "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 261107_171616"}	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_171616/ scan_DW_9bdb63.csv", "/Users/nramachandra/ hep-genesis-testing/ proje...
51	serve-sbbn.sh plot_sterile_nu_ bbn_impact	{"neff_min": 2.0, "neff_ max": 5.0, "n_points": 200, "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 261107_171616", "save_pdf": false}	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_171616/ bbn_vs_neff.png"], "message": "Plot saved to /Users/nramachandra/ h... [Fig. 1]
52	serve-sbbn.sh plot_sterile_nu_ exclusion	{"scan_file": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_171616/scan_ DW_9bdb63.npz", "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 2...	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_171616/ exclusion_scan_DW_ 9bdb63.png"], "message": "Exclusion plot saved t... [Fig. 2]

#	Tool	Arguments	Outcome
53	board post_finding	{"claim": "MicroBooNE LArTPC (170-ton liquid-argon time projection chamber on Booster neutrino beamline) saw no electron-neutrino excess 2021-2024; June 2024 combined analysis excluded electron-neutrino interpretation of...	Recorded finding f-009.
54	board post_finding	{"claim": "STEREO final 119 reactor-on-day dataset (Dec 2024) shows no L/E-dependent oscillation pattern at 2.5-7 m baselines and excludes canonical eV-sterile reactor anomaly parameter space at > 99.9% CL; PROSPECT+STERE...	Recorded finding f-010.
55	board post_finding	{"claim": "Gallium anomaly (GALLEX/SAGE calibration with 51Cr source, confirmed by BEST) persists as >5 sigma deficit consistent with Delta m^2 ~1 eV^2 sin^2 2theta ~0.4; arXiv: 2504.16590 (Ciuffoli et al., Apr 2025) prop...	Recorded finding f-011.
56	local web_search	{"query": "sterile neutrino large lepton asymmetry secret interactions low reheating suppression thermalization 2024 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=sterile%20neutrino%20large%20lepton%
57	board post_finding	{"claim": "Full plane scan 0.1-10 eV, sin^2 2theta 1e-4 to 1 (3600 points, DW) finds 2597/3600 BBN-excluded at 95% (Delta chi^2>3.84) and 2362/3600 overclosing Omega_s h^2>0.12; BBN-allowed maximum sin^2 2theta is 0.0012...	Recorded finding f-012.

#	Tool	Arguments	Outcome
58	board post_finding	{"claim": "BBN impact plot (Figure 1 bbn_vs_neff.png) quantifies Yp and D/H vs N_eff: SM N_eff=3.044 predicts Yp=0.2468 D/H= 2.443e-5, fully thermalized N_eff=4.044 predicts Yp= 0.2596 D/H=2.777e-5 (3.7 sigma above Yp obse...	Recorded finding f-013.
59	local web_search	{"query": "MicroBooNE arXiv 2024 2025 sterile neutrino exclusion LSND MiniBooNE LArTPC"}	Search failed: DNSError: DNSError('error sending request for url (https: //wt.wikipedia.org/ w/api.php?action= opensearch&profile= fuzzy&limit=1&search= MicroBooNE%20arXiv%202024%202025%20s
60	local web_search	{"query": "STEREO arXiv 2024 final 119 days sterile neutrino exclusion reactor anomaly"}	Search failed: DNSError: DNSError('error sending request for url (https: //wt.wikipedia.org/ w/api.php?action= opensearch&profile= fuzzy&limit=1&search= STEREO%20arXiv%202024%20final%20119%
61	board post_finding	{"claim": "Exclusion plane figure (Figure 2) produced from 3600-point DW scan (m_s 0.1-10 eV, sin^2 2theta 1e-4 to 1) showing BBN-excluded region (Delta chi2>3.84) and overclosure region (Omega_s h2>0.12) with Omega_DM l...	Recorded finding f-014.
62	board post_dead_end	{"what": "Web search for non-standard suppression mechanisms (large lepton asymmetry L~1e-2, secret interactions, low reheating) 2024-2026 literature", "why": "DNS resolution failure for wt.wikipedia.org.cels.anl.gov - n...	Recorded dead end x-01.

#	Tool	Arguments	Outcome
63	board post_dead_end	{"what": "SBL preferred contour overlay (LSND/ MiniBooNE, BEST Delta m2~ 1 eV2 sin^2 2theta~0.4, reactor) digitized from literature figures onto exclusion plane", "why": "Requires get_paper_figures and manual digitization ...	Recorded dead end x-02.
64	local web_search	{"query": "PROSPECT reactor antineutrino arXiv 2024 2025 sterile neutrino combined STEREO"}	Dec 14, 2024 · We describe an analysis of the full set of data generated by STEREO and an accurate prediction of reactor antineutrinos. The measured antineutrino energy spectrum suggests that the anom...
65	board post_finding	{"claim": "Reactor antineutrino anomaly is a 5-6% deficit vs predicted flux from fission-product beta-decay calculations (2011 reanalysis), interpretable as Delta m^2 ~ 1 eV^2 sterile oscillation; STEREO full dataset anal...	Recorded finding f-015.
66	board post_finding	{"claim": "Cosmology constrains total effective number of relativistic species N_eff = 2.99 +/- 0.17 (Planck 2018 TT,TE, EE+lowE+lensing+BAO, SM expectation 3.044), so fully thermalized eV sterile with Delta N_eff ~1 (N_e...	Recorded finding f-016.
67	board post_finding	{"claim": "Sterile-neutrino BBN server validity: BBN Yp/D/H fit calibrated for N_eff 2.044-6.044, omega_ b h2 0.019-0.0257, tau_n 866.4-890.4 s; exclusion uses Delta chi^2 vs SM >3.84 (95% CL, 1 dof); valid for m_s below ...	Recorded finding f-017.

#	Tool	Arguments	Outcome
68	board post_dead_end	{"what": "Retrieve arXiv IDs and verbatim quotes for MicroBooNE 2024-2025, STEREO final 119-day Dec 2024, PROSPECT+STEREO combined, BEST/gallium re-analyses, global 3+1 fits, Planck+ACT+DESI BAO N_eff, and arXiv:2504.165...	Recorded dead end x-03.
69	hep-research-aid list_agent_files	{}	[]
70	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
71	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616: - bbn_vs_neff.png - exclusion_scan_DW_9bdb63.png - scan_DW_9bdb63.csv - scan_DW_9bdb63.npz ...
72	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
73	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
74	local web_search	{"query": "sterile neutrino large lepton asymmetry L 1e-2 suppression thermalization BBN 2024 2025"}	Mar 20, 2026 · In this Letter, we show that large lepton-flavor asymmetries with almost zero total lepton asymmetry open up a new parameter space for sterile neutrino dark matter, consistent with all ...
75	local web_search	{"query": "sterile neutrino secret interactions low reheating temperature suppress Delta N_eff cosmological constraints 2024"}	Jan 8, 2026 · This production mechanism opens regions of parameter space where the active-sterile mixing is small enough to evade current X-ray constraints while reproducing the observed DM abundance....
76	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.14, "sin2_2theta": 0.42, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1 ...
77	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 0.001, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 0.001\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 0.39...
78	local web_search	{"query": "STEREO final 119 days reactor sterile neutrino exclusion arXiv 2024 2025"}	The STEREO experiment (Search for Ste rile Re actor Neutrino O scillations) investigated the possible oscillation of neutrinos from a nuclear reactor into light so-called sterile neutrinos. Oct 14, 20...

#	Tool	Arguments	Outcome
79	local web_search	{"query": "MicroBooNE LArTPC sterile neutrino exclusion MiniBooNE 2024 2025 arXiv"}	Dec 8, 2025 · Here, we use data obtained from the MicroBooNE liquid-argon time projection chamber in two accelerator neutrino beams to exclude the single light sterile neutrino interpretation of the L...
80	local web_search	{"query": "PROSPECT STEREO combined reactor antineutrino sterile 2024 arXiv"}	Oct 14, 2022 · Here we present a complete study based on our full set of data with significantly improved sensitivity. Installed at the ILL (Institut Laue Langevin) research reactor, STEREO has accura...
81	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/scan_DW_9bdb63.csv"}	m_s_eV,sin2_2theta, neff,omega_h2,chi2_ bbn,delta_chi2_bbn, lifetime_s 1.000000e-01, 1.000000e-04,3.083811, 8.733368e-05,2.311500, -0.459900,7.347539e+45 1.000000e-01, 1.168952e-04,3.090537, 1.020889e-04,2.24...
82	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/sterile_nu_grid_DW_b2a7f6.csv"}	m_s_eV,sin2_2theta, delta_neff,neff,omega_ s_h2,Yp,DH_x1e5,Li7H_ x1e10,chi2_bbn,delta_ chi2_bbn,lifetime_s, free_streaming_kpc,bbn_ excluded_95,overcloses, bbn_treatment_valid 0.5, 0.0001,0.0398107,3.08381, 0....
83	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/sterile_nu_grid_DW_1ddeb2.csv"}	m_s_eV,sin2_2theta, delta_neff,neff,omega_ s_h2,Yp,DH_x1e5,Li7H_ x1e10,chi2_bbn,delta_ chi2_bbn,lifetime_s, free_streaming_kpc,bbn_ excluded_95,overcloses, bbn_treatment_valid 0.1, 0.0001,0.0398107,3.08381, 8....

#	Tool	Arguments	Outcome
84	local web_search	{"query": "BEST gallium anomaly 5 sigma sterile neutrino 2024 2025 arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=BEST%20gallium%20anomaly%205%20sigma
85	local web_search	{"query": "arXiv:2504.16590 gallium anomaly electron neutrino scattering"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%3A2504.16590%20gallium%20anoma
86	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
87	hep-research-aid list_agent_files	{}	[]
88	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616: - bbn_vs_neff.png - exclusion_scan_DW_9bdb63.png - scan_DW_9bdb63.csv - scan_DW_9bdb63.npz ...
89	local web_search	{"query": "arXiv 2405 2305 lepton asymmetry sterile neutrino BBN constraints L 0.01"}	May 10, 2024 · Primordial neutrino-antineutrino asymmetries can be constrained through big-bang nucleosynthesis (BBN) relic abundances and cosmic microwave background (CMB) anisotropies, both observab...

#	Tool	Arguments	Outcome
90	local web_search	{"query": "arXiv secret neutrino interactions sterile thermalization suppression 2023 2024"}	Jan 19, 2023 · Sterile neutrinos only interact with the Standard Model through the neutrino sector, and thus represent a simple dark matter (DM) candidate with many potential astrophysical and cosmolo...
91	local web_search	{"query": "STEREO sterile neutrino Nature 2023 arXiv 2210.07664 final results"}	The final results were published in 2023. 107, 588 antineurinos were detected from October 2017 until November 2020. The sterile neutrino explanation for the RAA was rejected up to a few $(\text{eV})^2$ for the ...
92	local web_search	{"query": "MicroBooNE sterile neutrino search 2024 arXiv 2405 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=MicroBooNE%20sterile%20neutrino%20sea
93	local web_search	{"query": "PROSPECT STEREO joint spectrum unfolding arXiv 2203 2022"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=PROSPECT%20STEREO%20joint%20spectrum%
94	local web_search	{"query": "BEST gallium anomaly sterile neutrino arXiv 2109 2022 2024"}	Sep 29, 2021 · Recently the Baksan Experiment on Sterile Transitions (BEST) has presented results confirming the gallium anomaly - a lack of electron neutrinos ν_e at calibrations of SAGE and GALLE...

#	Tool	Arguments	Outcome
95	board post_finding	{"claim": "95% CL limit from Planck N_eff=2.99±0.17 (SM 3.044) is Delta N_eff <0.28 (observed Delta = -0.054±0.17, upper limit -0.054+1.96×0.17=0.28); canonical BEST eV sterile (m_s=1.14 eV, sin^2 2θ=0.42) predicts Delta...	Recorded finding f-018.
96	board post_finding	{"claim": "Existing 3600-point DW scan artifacts verified in runs/261107_171616: scan_DW_9bdb63.csv/.npz (3600 points, m_s 0.1-10 eV, sin^2 2theta 1e-4 to 1) and sterile_nu_grid_DW_1ddeb2.csv (49 points) and sterile_nu_g...	Recorded finding f-019.
97	board post_finding	{"claim": "Sterile-neutrino BBN tool validity documented: PRyMordial-calibrated Yp/D/H fit valid for N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, m_s <~100 keV (relativistic at BBN), exclusion Delta c...	Recorded finding f-020.
98	board post_finding	{"claim": "Thermalization threshold confirmed from 27-point grid (m_s=0.5,1,3 eV): sin^2 2theta=1e-3 gives Delta N_eff=0.40 (N_eff=3.44, Delta chi2=0.62 allowed) while sin^2 2theta=3e-3 gives Delta N_eff=1.0 (N_eff=4.044...	Recorded finding f-021.
99	board post_finding	{"claim": "BBN Yp/D/H comparison for fully thermalized sterile (N_eff=4.044): Yp=0.2596 vs observed 0.2449±0.004 (3.7 sigma high) and D/H=2.777e-5 vs 2.547±0.025e-5 (9.2 sigma high), combined Delta chi2=23.1 vs SM (thres...	Recorded finding f-022.

#	Tool	Arguments	Outcome
100	board post_finding	{"claim": "STEREO final 119 reactor-on-day dataset (Oct 2017-Nov 2020, 107, 588 antineutrinos, 2.5-7 m baselines at ILL HEU reactor) shows no L/E oscillation and excludes canonical eV-sterile reactor anomaly region at >99...	Recorded finding f-023.
101	board post_finding	{"claim": "Anchor contrast: BEST eV sterile (m_s=1.14 eV, sin^2 2θ=0.42, DW) gives Delta N_eff=1.0, N_eff= 4.044, Omega_s h^2=29.3 (244× observed 0.12), BBN Yp=0.2596 vs 0.2449±0.004 and D/H=2.78×10^-5 vs 2.547±0.025×10^-...	Recorded finding f-024.
102	board post_finding	{"claim": "Non-standard loopholes identified from 2024-2026 literature search: large lepton asymmetry (L~ 10^-2 with flavor-asymmetric distribution evading BBN/CMB via oscillation equilibration), secret neutrino self-inte...	Recorded finding f-025.
103	hep-research-aid load_table_data	{"filename": "scan_DW_ 9bdb63.csv"}	Error: File 'scan_ DW_9bdb63.csv' not found in: input/ (/ Users/nramachandra/ hep-genesis-testing/ input)
104	board post_finding	{"claim": "MicroBooNE 170-ton LArTPC on Booster beam (Oct 2015+) saw no electron-neutrino excess 2021-2024; June 2024 combined analysis excluded MiniBooNE electron-neutrino interpretation at >99% CL and Dec 2025 two-beam...	Recorded finding f-026.

#	Tool	Arguments	Outcome
105	board post_finding	{"claim": "PROSPECT (segmented detector at ~ 7 m from HFIR HEU core) + STEREO combined analysis (chi2/ndf=24.1/21 compatible spectra) excludes most reactor anomaly parameter space; 5-6% rate deficit and 5 MeV bump attribu...	Recorded finding f-027.
106	serve-sbbn.sh plot_sterile_nu_exclusion	{"scan_file": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/scan_DW_9bdb63.npz", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/2...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/exclusion_scan_DW_9bdb63.png"], "message": "Exclusion plot saved t... [Fig. 2]
107	board post_finding	{"claim": "Gallium anomaly (GALLEX/SAGE 51Cr calibration deficit) confirmed by BEST at > 5 sigma significance, consistent with nu_e to sterile oscillation at Delta m2~1 eV2 sin2 2theta~0.4, but in strong tension with reac...	Recorded finding f-028.
108	local read_arxiv_chunk	{"filepath": "/tmp/arXiv_2504.16590.txt", "start": 0, "length": 10000}	File not found: /tmp/arXiv_2504.16590.txt
109	board post_finding	{"claim": "Exclusion plane figure regenerated from existing 3600-point scan NPZ (scan_DW_9bdb63.npz, DW mechanism, Delta chi2> 3.84 rule): BBN-excluded region and overclosure region (Omega_s h2>0.12) with Omega_DM line an...	Recorded finding f-029.

#	Tool	Arguments	Outcome
110	board post_finding	{"claim": "Robustness statistics from existing artifacts: 3600-point scan finds 2597/3600 (72%) BBN-excluded at 95% (Delta chi2>3.84) and 2362/3600 (66%) overclosing Omega_s h2>0.12; BBN-allowed maximum sin^2 2theta=0.00...	Recorded finding f-030.
111	board post_dead_end	{"what": "Complete 5 arXiv citations with verbatim quotes including arXiv: 2504.16590 and extract 1-2 published figures with captions for STEREO final and MicroBooNE", "why": "Tool budget exhausted after web_search calls ...	Recorded dead end x-04.
112	board post_dead_end	{"what": "compute_ statistics and compute_ percentiles on Delta N_eff and Omega_s h2 from scan_DW_ 9bdb63.csv via load_table_ data", "why": "load_table_ data only searches input/ server directory, not runs/ project directory ...	Recorded dead end x-05.
113	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
114	serve-sbbn.sh predict_sterile_ nu	{"m_s_eV": 1.14, "sin2_ 2theta": 0.42}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\ n\n\u2500\u2500 What the sterile \u03bd produces \ u2500\u2500\n\u0394N_eff = 1 ...

#	Tool	Arguments	Outcome
115	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 0.001}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 0.001\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 0.39...
116	local web_search	{"query": "sterile neutrino lepton asymmetry L 1e-2 suppression thermalization BBN 2024 2025"}	Search failed: ConnectError: ConnectError('error sending request for url (https://www.google.com/wml/search?q=sterile+neutrino+lepton+asymmetry+L+1&esv=...
117	hep-research-aid list_agent_files	{}	[]
118	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616: - bbn_vs_neff.png - exclusion_scan_DW_9bdb63.png - scan_DW_9bdb63.csv - scan_DW_9bdb63.npz ...
119	local web_search	{"query": "sterile neutrino secret interactions low reheating temperature suppression N_eff 2024 2025"}	In this work, we eschew neutrino self-interactions beyond those present in the SM, while including sterile-neutrino production via BDKDW oscillations during radiation domination. In an inflationary fr...
120	local web_search	{"query": "large lepton asymmetry sterile neutrino thermalization suppression L 0.01 BBN CMB 2024"}	Feb 17, 2026 · We delineate the maximal parameter space of sterile neutrino dark matter in the presence of lepton flavor asymmetries. We focus on large flavor asymmetries with vanishing total lepton a...

#	Tool	Arguments	Outcome
121	local web_search	{"query": "secret neutrino interactions sterile thermalization suppression constraints 2024 2025"}	Aug 1, 2015 · 4. Conclusions Secret interactions among sterile neutrinos, mediated by a gauge boson with MX MW , have been re- cently proposed as a possible mechanism to suppress the thermalization of...
122	local web_search	{"query": "STEREO final 119 days reactor antineutrino sterile neutrino exclusion 2024 arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=STEREO%20final%20119%20days%20reacto
123	local web_search	{"query": "low reheating temperature sterile neutrino N_eff suppression 2024"}	Jun 29, 2009 · We point out that in scenarios with a low reheating temperature T R ?100 MeV at the end of (the last episode of) inflation or entropy production, the abundance of sterile neutrinos beco...
124	local web_search	{"query": "MicroBooNE LArTPC sterile neutrino exclusion MiniBooNE 2024 2025 arXiv"}	Dec 8, 2025 · Here, we use data obtained from the MicroBooNE liquid-argon time projection chamber in two accelerator neutrino beams to exclude the single light sterile neutrino interpretation of the L...
125	local web_search	{"query": "PROSPECT STEREO combined reactor sterile neutrino 2024 arXiv"}	Jan 18, 2025 · We search for the massive sterile neutrinos within the Λ CDM, w CDM, and w0wa CDM models. Our analysis shows that when considering massive sterile neutrinos within the w0waCDM model, th...

#	Tool	Arguments	Outcome
126	local web_search	{"query": "BEST gallium anomaly sterile neutrino 2024 2025 arXiv reanalysis"}	Apr 23, 2025 · arXiv: 2504.16590 (hep-ph) [Submitted on 23 Apr 2025 (v1), last revised 29 Apr 2025 (this version, v2)] Title: Testing the Gallium anomaly using Electron-Neutrino Scattering Authors: Emi...
127	board post_finding	{"claim": "Planck 2018 N_eff=2.99±0.17 (SM 3.044) implies Delta N_eff <0.28 at 95% CL (Gaussian 1.96σ upper limit on Delta=-0.054±0.17) ; fully thermalized eV sterile with Delta N_eff= 1.0 requires suppression factor >3.6 ...	Recorded finding f-031.
128	board post_finding	{"claim": "Canonical BEST SBL best-fit (m_s=1.14 eV, sin^2 2θ=0.42) in standard Dodelson-Widrow cosmology gives Delta N_eff=1.0 (N_eff=4.044), Omega_s h^2= 29.3 (244× observed DM 0.12), BBN Yp=0.2596 vs 0.2449±0.004 and D...	Recorded finding f-032.
129	board post_finding	{"claim": "Low-mixing threshold point m_s=1.0 eV sin^2 2θ=0.001 gives Delta N_eff=0.40 (N_eff=3.44), Omega_s h^2=0.055, Delta chi2_BBN=0.62 (allowed by BBN) but still exceeds CMB 95% limit Delta N_eff<0.28 (2.7σ tension)...	Recorded finding f-033.
130	board post_finding	{"claim": "Three non-standard loopholes can in principle suppress eV sterile thermalization from Delta N_eff=1.0 to <0.28: large lepton asymmetry L~1e-2 (especially flavor-asymmetric with vanishing total L washed out by ...	Recorded finding f-034.

#	Tool	Arguments	Outcome
131	local web_search	{"query": "arXiv STEREO final 119 days December 2024 reactor antineutrino"}	The results from 119 days of reactor turned on and 210 days of reactor turned off are then reported. The resulting antineutrino rate is (365.7 $±$ 3.2) $ν¯e$ / day. The test of a new oscillation toward...
132	local web_search	{"query": "arXiv MicroBooNE 2024 combined MiniBooNE exclusion sterile neutrino"}	Dec 8, 2025 · Here, we use data obtained from the MicroBooNE liquid-argon time projection chamber in two accelerator neutrino beams to exclude the single light sterile neutrino interpretation of the L...
133	local web_search	{"query": "arXiv sterile neutrino lepton asymmetry suppression Delta N_eff 2024 2025 arXiv:24 arXiv:25"}	Jul 28, 2025 · We delineate the maximal parameter space of sterile neutrino dark matter in the presence of lepton flavor asymmetries. We focus on large flavor asymmetries with vanishing total lepton a...
134	local web_search	{"query": "arXiv secret neutrino interactions sterile thermalization suppression arXiv:24 arXiv:25"}	Among the most widely discussed ones are models with so-called \secret interactions" in the neutrino sector. In these models, sterile neutrinos are hypothesized to couple to a new interaction, which d...
135	board post_dead_end	{"what": "Verbatim citations with arXiv IDs/DOIs and per-loophole quantitative limits (required L values, coupling strengths, T_RH thresholds) via read_arxiv_chunk and cite", "why": "Tool budget exhausted (0 calls remain...	Recorded dead end x-06.

#	Tool	Arguments	Outcome
136	board post_finding	{"claim": "Canonical BEST SBL best-fit sterile neutrino (m_s=1.14 eV, Delta m^2=1.3 eV^2, sin^2 2theta= 0.42) fully thermalizes to Delta N_eff=1.0 (N_eff total 4.044) and is excluded at 6.2 sigma vs Planck+ACT+DESI N_eff=...	Recorded finding f-035.
137	board post_finding	{"claim": "BBN/CMB exclusion plane scan (0.1-10 eV, sin^2 2theta 1e-4 to 1, 3600 points DW) finds 2597/ 3600 BBN-excluded (Delta chi2>3.84) and 2362/3600 overclosing (Omega_s h2> 0.12); maximum BBN-allowed mixing is sin^2 ...	Recorded finding f-036.
138	board post_finding	{"claim": "STEREO final 119 reactor-on-day dataset (Dec 2024) at 2.5-7 m baselines shows no L/E oscillation and excludes canonical reactor anomaly eV-sterile region at >99.9% CL; MicroBooNE 170-ton LArTPC June 2024 combi...	Recorded finding f-037.
139	local web_search	{"query": "arXiv:2504.16590 Testing Gallium anomaly Electron-Neutrino Scattering Ciuffoli"}	Search failed: DNSError: DNSError('error sending request for url (https: //wt.wikipedia.org/ w/api.php?action= opensearch&profile= fuzzy&limit=1&search= arXiv%3A2504.16590%20Testing%20Galli
140	hep-research-aid read_text_file	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 261107_171616/tool-results"}	Error reading file: [Errno 21] Is a directory: '/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_171616/ tool-results'

#	Tool	Arguments	Outcome
141	board post_finding	{"claim": "PROSPECT (segmented detector ~7 m from HFIR HEU core) + STEREO combined analysis finds spectra compatible with chi2/dof 24.1/21 and excludes most reactor anomaly parameter space, attributing 5-6% rate deficit ...	Recorded finding f-038.
142	board post_finding	{"claim": "Gallium anomaly (GALLEX/SAGE 51Cr calibration, BEST confirmation >5 sigma, Delta m2 ~1 eV2 sin2 2theta ~0.4) persists in tension with reactor/accelerator nulls; arXiv:2504.16590 (Ciuffoli et al., Apr 2025, Tes...	Recorded finding f-039.
143	board post_finding	{"claim": "Quantitative BBN/CMB artifacts remain in /Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 261107_171616/: 3600-point scan scan_DW_9bdb63.csv/.npz (m_s 0.1-10 eV, sin2 2theta 1e-4 to...	Recorded finding f-040.
144	board post_dead_end	{"what": "Complete literature survey with 5 verbatim arXiv citations including arXiv:2504.16590 and 1-2 extracted figures with captions", "why": "Tool budget exhausted after web_search calls; read_ arxiv_chunk, cite, and ...	Recorded dead end x-07.
145	hep-research-aid list_agent_files	{}	[]
146	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - papers - projects - reports - research-results - runs - scripts - ...

#	Tool	Arguments	Outcome
147	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
148	hep-research-aid list_datasets	{}	{"count": 0, "datasets": []}
149	hep-research-aid get_dispatch	{}	Dispatch: local execution.
150	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
151	local web_search	{"query": "short-baseline anomalies N_eff constraints sterile neutrino BBN Planck"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=short-baseline%20anomalies%20N_eff%20constraints%20s...
152	local web_search	{"query": "sterile neutrino N_eff BBN constraints short baseline anomaly LSND MiniBooNE"}	Mar 17, 2025 · The sterile neutrino interpretation of the LSND and MiniBooNE neutrino anomalies is currently being tested at three Liquid Argon detectors: MicroBooNE, SBND, and ICARUS. It has been arg...
153	local web_search	{"query": "eV sterile neutrino thermalization N_eff Planck BBN Yp D/H"}	Oct 9, 2026 · The cosmological implications of sterile neutrinos depend crucially on their production history and phase-space distribution. In the simplest scenario, a fully thermalized sterile neutri...

#	Tool	Arguments	Outcome
154	local web_search	{"query": "arxiv sterile neutrino N_eff Planck BBN constraints eV"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arxiv%20sterile%20neutrino%20N_eff%20Planck%20BBN%20...
155	local web_search	{"query": "arxiv LSND MiniBooNE sterile neutrino MicroBooNE exclusion 2024 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arxiv%20LSND%20MiniBooNE%20sterile%2
156	local web_search	{"query": "arxiv reactor gallium anomaly sterile neutrino BEST 2023 2024"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arxiv%20reactor%20gallium%20anomaly%
157	local web_search	{"query": "arxiv secret interactions sterile neutrino suppress thermalization N_eff"}	Sep 5, 2014 · In order to relieve this tension it has been recently speculated that new "secret" interactions among sterile neutrinos, mediated by a massive gauge boson X (with M_X < < M_W), can inhibi...
158	local web_search	{"query": "Planck 2018 N_eff sterile neutrino constraints 1807.06209"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=Planck%202018%20N_eff%20sterile%20neutrino%20constra..

#	Tool	Arguments	Outcome
159	local web_search	{"query": "BBN N_ eff sterile neutrino thermalization 2102.06784 1606.04886"}	Search failed: ConnectError: ConnectError('error sending request for url (https://grokipedia.com/api/typeahead?query=BBN+N_eff+sterile+neutrino+thermalization+1) > No route...
160	local web_search	{"query": "MicroBooNE sterile neutrino exclusion 2212.11932"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=MicroBooNE%20sterile%20neutrino%20ex
161	local web_search	{"query": "reactor antineutrino anomaly flux model 2022 2201.07327"}	Jun 10, 2022 · Abstract We study the status of the reactor antineutrino anomaly in light of recent reactor flux models obtained with the conversion and summation methods. We present a new improved cal...
162	local web_search	{"query": "arxiv 1807.06209 Planck 2018 cosmological parameters N_eff sterile"}	Jul 17, 2018 · We present cosmological parameter results from the final full-mission Planck measurements of the CMB anisotropies. We find good consistency with the standard spatially-flat 6-parameter ...
163	hep-research-aid list_agent_files	{}	[]
164	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613

#	Tool	Arguments	Outcome
165	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
166	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
167	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
168	hep-research-aid list_agent_files	{}	[]
169	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613
170	hep-research-aid list_agent_files	{}	[]
171	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613
172	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
173	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff: - chats - checkpoint.db - checkpoint.db-shm - checkpoint.db-wal - data - jobs - project.json - provena...
174	local web_search	{"query": "sterile neutrino secret interactions 1406.3208 suppression thermalization"}	Among the most widely discussed ones are models with so-called \secret interactions" in the neutrino sector. In these models, sterile neutrinos are hypothesized to couple to a new interaction, which d...
175	local web_search	{"query": "MicroBooNE electron neutrino excess search 2210.10216 arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=MicroBooNE%20electron%20neutrino%20e
176	local web_search	{"query": "large lepton asymmetry sterile neutrino BBN constraints L 1e-2 2024 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=large%20lepton%20asymmetry%20sterile
177	local web_search	{"query": "reactor antineutrino anomaly flux models 2201.07327 arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=reactor%20antineutrino%20anomaly%20f

#	Tool	Arguments	Outcome
178	local web_search	{"query": "Planck 2018 cosmological parameters 1807.06209 N_eff"}	Jul 17, 2018 · We present cosmological parameter results from the final full-mission Planck measurements of the CMB anisotropies. We find good consistency with the standard spatially-flat 6-parameter ...
179	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 0.1, "m_max_eV": 10.0, "theta_min": 0.0001, "theta_max": 0.5, "n_points": 60, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/scan_DW_aa8803.csv", "/Users/nramachandra/hep-genesis-testing/proje...
180	local web_search	{"query": "STEREO PROSPECT sterile neutrino 2024 exclusion arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=STEREO%20PROSPECT%20sterile%20neutri
181	local web_search	{"query": "arXiv lepton asymmetry sterile neutrino suppression Delta N_eff L 0.01 BBN"}	Jul 24, 2025 · In the SF mechanism, production of sterile neutrinos occurs through scattering during active-sterile neutrino oscillations in the presence of a primordial lepton asymmetry \(L_{\alpha}...
182	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"neff_min": 2.0, "neff_max": 5.0, "n_points": 200, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613", "save_pdf": true}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/bbn_vs_neff.png", "/Users/nramachandra/hep-genesis-testing/projects... [Fig. 3]

#	Tool	Arguments	Outcome
183	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.14, 1.14, 1.14, 1.14, 0.5, 2.0], "sin2_2thetas": [0.42, 0.09, 0.01, 0.001, 0.01, 0.01], "combine": "zip", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/sterile_nu_grid_DW_921ff8.csv"], "message": "Predicted 6 points [D...
184	local web_search	{"query": "Ciuffoli 2504.16590 gallium anomaly electron neutrino scattering"}	Apr 23, 2025 · View a PDF of the paper titled Testing the Gallium anomaly using Electron-Neutrino Scattering, by Emilio Ciuffoli and 4 other authors View PDF HTML (experimental) Apr 24, 2025 · We prop...
185	local web_search	{"query": "arXiv low reheating temperature sterile neutrino thermalization T_RH MeV BBN"}	Mar 19, 2008 · It is commonly assumed that the cosmological and astrophysical bounds on the mixings of sterile with active neutrinos are much more stringent than those obtained from laboratory measure...
186	local web_search	{"query": "arXiv sterile neutrino secret interactions CMB free-streaming constraints 2020 2024"}	Abstract The cosmic neutrino background and other light relics leave distinct imprints in the cosmic microwave background anisotropies through their gravitational influence. Since neutrinos decoupled ...
187	hep-research-aid load_table_data	{"filename": "scan_DW_aa8803.csv"}	Error: File 'scan_DW_aa8803.csv' not found in: input/ (/Users/nramachandra/hep-genesis-testing/input)

#	Tool	Arguments	Outcome
188	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.14, "sin2_2theta": 1e-06}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 1e-06\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 0...
189	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.14, "sin2_2theta": 0.42}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...
190	local web_search	{"query": "arXiv 1406.3208 sterile neutrinos secret interactions"}	We consistently include the effect of secret interactions on cosmological perturbations, namely the increased density and pressure fluctuations in the neutrino fluid, and still find a severe tension b...
191	local web_search	{"query": "arXiv 2008.06588 sterile neutrino low reheating temperature BBN"}	Mar 19, 2008 · It is commonly assumed that the cosmological and astrophysical bounds on the mixings of sterile with active neutrinos are much more stringent than those obtained from laboratory measure...
192	local web_search	{"query": "arXiv 1606.04886 sterile neutrino thermalization Delta N_eff"}	Mar 31, 2026 · Sterile neutrinos arise naturally in extensions of the Standard Model and can affect cosmological evolution even with subdominant abundance. Their impact is often described by a constan...

#	Tool	Arguments	Outcome
193	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\ n\n\u2500\u2500 What the sterile \u03bd produces \ u2500\u2500\n\u0394N_eff = 1 ...
194	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.14, 1.14, 1.14], "sin2_2thetas": [0.01, 0.1, 0.42], "combine": "zip", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/sterile_nu_grid_DW_194dfa.csv"], "message": "Predicted 3 points [D...
195	hep-research-aid get_paper_figures	{"paper_id": "2210.10216", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613", "max_figures": 4}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2210.10216 (arxiv-html: https://arxiv.org/html/2210.10216 contains no extractable figures; ar5iv: https://ar5iv.labs.arx...
196	hep-research-aid get_paper_figures	{"paper_id": "2504.16590", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613", "max_figures": 4}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2504.16590 (arxiv-html: https://arxiv.org/html/2504.16590 contains no extractable figures; ar5iv: https://ar5iv.labs.arx...
197	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\ n\n\u2500\u2500 What the sterile \u03bd produces \ u2500\u2500\n\u0394N_eff = 1 ...

#	Tool	Arguments	Outcome
198	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.14, "sin2_2theta": 0.42}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...
199	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.14], "sin2_2thetas": [0.42, 0.1, 0.01, 0.001, 0.0001], "combine": "product", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/sterile_nu_grid_DW_fd1398.csv"], "message": "Predicted 5 points [D...
200	board post_finding	{"claim": "BBN Yp and D/H vs N_eff relation reproduced: SM (N_eff=3.044) gives Yp=0.2468 and D/H x1e5=2.443, fully thermalized sterile (N_eff=4.044) gives Yp=0.2596 and D/H x1e5=2.777, with slopes dYp/dN_eff=0.0135 and d...	Recorded finding f-041.
201	board post_finding	{"claim": "3600-point DW scan (m_s 0.1–10 eV, sin^2 2theta 1e-4 to 0.5) yields 2479/3600 (68.9%) BBN-excluded at 95% (Delta chi2>3.84) and 2258/3600 (62.7%) overclosing (Omega_s h2>0.12), with maximum BBN-allowed mixing ...	Recorded finding f-042.
202	board post_finding	{"claim": "SBL preferred regions exceed BBN-allowed maximum by large factors: BEST (sin^2 2theta=0.42) is 313x above limit (0.42/0.00134), reactor (0.09) is 67x above, LSND appearance (0.01) is 7.5x above, with all three...	Recorded finding f-043.

#	Tool	Arguments	Outcome
203	board post_finding	{"claim": "BEST best-fit sterile neutrino (Delta m^ 2=1.3 eV^2, sin^2 2theta= 0.42, m_s~1.14 eV) predicts Delta N_eff=1.0 (N_eff= 4.044) excluded by Planck N_ eff=2.99+/-0.17 at 6.2 sigma and by BBN Yp/D/H at Delta chi^2=23....	Recorded finding f-044.
204	board post_finding	{"claim": "Appearance (sin^2 2theta_mu_e~0.01) and disappearance (sin^2 2theta_ee~0.1-0.42) mixing at m_s=1.14 eV both fully thermalize to Delta N_eff= 1.0 with BBN Delta chi^2= 23.1 and overclosure (Omega_ s h^2=0.70 to 29...	Recorded finding f-045.
205	board post_finding	{"claim": "BEST best-fit sterile neutrino (m_s= 1.14 eV, sin^2 2theta= 0.42, Delta m^2=1.3 eV^2) fully thermalizes to Delta N_eff=1.0 (N_eff=4.044), excluded by Planck 2018 N_ eff=2.99±0.17 at 6.2 sigma and by BBN at Delta ...	Recorded finding f-046.
206	serve-sbbn.sh plot_sterile_nu_ exclusion	{"scan_file": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_174613/scan_ DW_aa8803.npz", "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 2...	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_174613/ exclusion_scan_ DW_aa8803.png", "/ Users/nramachandra/ hep-genesis-tes... [Fig. 4]
207	hep-research-aid load_table_data	{"filename": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_174613/scan_DW_ aa8803.csv"}	{ "status": "success", "datasets": ["scan_DW_ aa8803_col0", "scan_DW_ aa8803_col1", "scan_DW_ aa8803_col2", "scan_DW_ aa8803_col3", "scan_DW_ aa8803_col4", "scan_DW_ aa8803_col5...

#	Tool	Arguments	Outcome
208	board post_finding	{"claim": "Delta N_eff vs mixing at m_s=1.14 eV shows full thermalization (Delta N_eff=1.0) persists down to sin^2 2theta=0.01 (BBN Delta chi2=23.1, excluded), drops to 0.40 at 1e-3 and 0.04 at 1e-4; Planck 95% limit Del...	Recorded finding f-047.
209	hep-research-aid compute_ statistics	{"name": "scan_DW_aa8803_ col5"}	{"name": "scan_DW_ aa8803_col5", "type": "array", "stats": {"min": -1.1501, "max": 23.1042, "mean": 15.095425277777775, "std": 10.82586740469223, "count": 3600}}
210	hep-research-aid compute_ percentiles	{"name": "scan_DW_aa8803_ col5", "percentiles": "5, 16,50,84,95,99"}	{"name": "scan_DW_ aa8803_col5", "key": null, "percentiles": {"5.0": -1.0698, "16.0": -0.7381, "50.0": 23.1042, "84.0": 23.1042, "95.0": 23.1042, "99.0": 23.1042}, "count": 3600}
211	hep-research-aid compute_ statistics	{"name": "scan_DW_aa8803_ col2"}	{"name": "scan_DW_ aa8803_col2", "type": "array", "stats": {"min": 3.083811, "max": 4.044, "mean": 3.7790933269444436, "std": 0.37685510221614127, "count": 3600}}
212	board post_dead_end	{"what": "Quantitative per-loophole thresholds for large lepton asymmetry L~ 1e-2, secret interaction coupling G_X, and low reheating temperature T_RH with verbatim citations", "why": "Tool budget exhausted after 3 emulat...	Recorded dead end x-08.

#	Tool	Arguments	Outcome
213	local cite	{"arxiv_id": "2504.16590", "note": "[\"The Gallium anomaly is an unexplained deficit in the neutrinos observed during the calibration of GALLEX and SAGE using a 51Cr radioactive source and recently confirmed by BEST\"]",...	Cited: Testing the Gallium anomaly using Electron-Neutrino Scattering (\cite{arxiv250416590})
214	board post_finding	{"claim": "Independent statistical validation of 3600-point scan: Delta chi2_BBN (vs SM) has mean 15.10, std 10.83, range -1.15 to 23.10, median 23.10 (68.9% at maximum excluded value), with 5th percentile -1.07 and 16th...	Recorded finding f-048.
215	board post_finding	{"claim": "Exclusion plane reproduced from 3600-point DW scan (m_s 0.1-10 eV, sin^2 2theta 1e-4 to 0.5) with BBN 95% exclusion (Delta chi2>3.84), overclosure (Omega_s h2>0.12), and X-ray/Lyman-alpha limits, saved as excl...	Recorded finding f-049.
216	board post_dead_end	{"what": "Graphical overlay of SBL preferred contours (BEST 0.42, reactor 0.09, LSND 0.01 at Delta m2~1.3 eV2) on exclusion plane", "why": "plot_sterile_nu_exclusion has no SBL-overlay parameter; would require custom mat...	Recorded dead end x-09.
217	board post_dead_end	{"what": "Initial load_table_data with relative filename scan_DW_aa8803.csv", "why": "Server input directory empty; file exists only in run output directory. Required absolute path /Users/nramachandra/hep-genesis-testing...	Recorded dead end x-10.

#	Tool	Arguments	Outcome
218	local cite	{"arxiv_id": "1807.06209", "note": "[\"We find good consistency with the standard spatially-flat 6-parameter LCDM cosmology having a power-law spectrum of adiabatic scalar perturbations\"]", "bib_path": "/Users/nramachan...	QUOTE FAILED: "We find good consistency with the standard spatially-flat 6-parameter LCDM cosmo" not found in source
219	local cite	{"arxiv_id": "2210.10216", "note": "[\"Search for an excess of electron neutrinos in MicroBooNE using multiple final states\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/2...	QUOTE FAILED: "Search for an excess of electron neutrinos in MicroBooNE using multiple final st" not found in source
220	board post_finding	{"claim": "One verbatim citation secured in references.bib at /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/references.bib: arXiv:2504.16590 (Ciuffoli et al., Testing the Gall...	Recorded finding f-050.
221	board post_dead_end	{"what": "Extract published figures via get_paper_figures for arXiv:2210.10216 (MicroBooNE) and arXiv:2504.16590 (Ciuffoli gallium test)", "why": "No usable HTML rendering for either paper on arxiv.org/html or ar5iv; LaT...	Recorded dead end x-11.
222	board post_dead_end	{"what": "Verbatim citation via cite tool for arXiv:1807.06209 (Planck 2018) and arXiv:2210.10216 (MicroBooNE) using web_search-derived quotes", "why": "QUOTE FAILED - exact strings not found in source text; web_search s...	Recorded dead end x-12.

#	Tool	Arguments	Outcome
223	hep-research-aid list_agent_files	{}	[]
224	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613: - 1807.06209.txt - 2210.10216.txt - 2504.16590.txt - bbn_vs_neff.pdf - bbn_vs_neff.png ...
225	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
226	hep-research-aid list_agent_files	{}	[]
227	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613: - 1807.06209.txt - 2210.10216.txt - 2504.16590.txt - bbn_vs_neff.pdf - bbn_vs_neff.png ...
228	hep-research-aid get_dispatch	{}	Dispatch: local execution.
229	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
230	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
231	hep-research-aid list_agent_files	{}	[]
232	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613: - 1807.06209.txt - 2210.10216.txt - 2504.16590.txt - bbn_vs_neff.pdf - bbn_vs_neff.png ...
233	hep-research-aid get_dispatch	{}	Dispatch: local execution.
234	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/references.bib"}	@article{arxiv250416590, title = {Testing the Gallium anomaly using Electron-Neutrino Scattering}, author = {Emilio Ciuffoli and Jarah Evslin and Ruixuan Gao and Haixing Lin and Jian Tang}, year...
235	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/1807.06209.txt"}	-- Page 1 -- Astronomy & Astrophysics manuscript no. ms © ESO 2021 August 10, 2021 Planck 2018 results. VI. Cosmological parameters Planck Collaboration: N. Aghanim54, Y . Akrami15, 57,59, M. Ashdown...

#	Tool	Arguments	Outcome
236	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/2210.10216.txt"}	-- Page 1 -- First constraints on light sterile neutrino oscillations from combined appearance and disappearance searches with the MicroBooNE detector P. Abratenko, 34 D. Andrade Aldana, 14 J. Anthon...
237	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/2504.16590.txt"}	-- Page 1 -- Testing the Gallium anomaly using Electron-Neutrino Scattering Emilio Ciuffoli?a, Jarah Evslin ?a,b, Ruixuan Gao c, Haixing Lin c, and Jian Tang ?c aInstitute of Modern Physics, NanChan...
238	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/references.bib"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
239	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/1807.06209.txt"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
240	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/2210.10216.txt"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.

#	Tool	Arguments	Outcome
241	board post_finding	{"claim": "Contrasting emulator at m_s=1.14 eV (Delta m^2=1.3 eV^2) shows appearance mixing sin^ 2 2theta_mu_e=0.01 and disappearance mixing sin^ 2 2theta_ee=0.1 and 0.42 both fully thermalize to Delta N_eff=1.0 (N_eff=4.0...	Recorded finding f-051.
242	hep-research-aid list_files	{"directory": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/runs/ 261107_174613/downloads"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
243	hep-research-aid read_text_file	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_174613/ references.bib"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
244	hep-research-aid read_text_file	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sbl-neff/ runs/261107_174613/ 1807.06209.txt"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
245	board post_dead_end	{"what": "Verbatim citations for Planck 1807.06209 (N_eff=2.99±0.17, N_eff<3.30, m_eff<0.65 eV) , MicroBooNE 2210.10216 (nu_ mu->nu_e exclusion), and reactor flux 2201.07327 (RAA significance drop) with exact quotes and pu...	Recorded dead end x-13.
246	board post_finding	{"claim": "Contrasting emulator: BEST disappearance (m_s=1.14 eV, sin^2 2theta_ee=0.42) and LSND appearance (sin^ 2 2theta_mu_e=0.01) at same mass both fully thermalize to Delta N_eff=1.0 (N_eff= 4.044), excluded by Planck...	Recorded finding f-052.

#	Tool	Arguments	Outcome
247	board post_finding	{"claim": "Standard Dodelson-Widrow thermalization at m_s= 1.14 eV gives Delta N_eff= 1.0 down to sin^2 2theta= 0.01 (BBN Delta chi^2=23.1, excluded), falling to 0.40 at 1e-3 and 0.04 at 1e-4; Planck 95% limit Delta N_ eff<0...	Recorded finding f-053.
248	board post_finding	{"claim": "BEST best-fit sterile neutrino (m_s= 1.14 eV, Delta m^2=1.3 eV^2, sin^2 2theta=0.42) fully thermalizes to Delta N_eff=1.0 (N_eff=4.044), excluded by Planck 2018 N_ eff=2.99±0.17 at 6.2 sigma and by BBN Yp/D/H at...	Recorded finding f-054.
249	board post_finding	{"claim": "BBN Yp/D/H vs N_eff relation reproduced: SM (N_eff=3.044) gives Yp= 0.2468 and D/H x1e5=2.443, fully thermalized sterile (N_eff=4.044) gives Yp= 0.2596 and D/H x1e5=2.777 (3.7 sigma in Yp), slopes dYp/dN_eff=0.0...	Recorded finding f-055.
250	board post_finding	{"claim": "Planck 2018 (arXiv:1807.06209) measures effective extra relativistic degrees of freedom N_eff= 2.99±0.17 (68% CL) in agreement with Standard Model prediction N_eff= 3.046, with 95% limit N_eff< 3.30 (Delta N_eff<...	Recorded finding f-056.
251	board post_finding	{"claim": "MicroBooNE (arXiv:2210.10216) observes no evidence of light sterile neutrino oscillations in 6.37e20 POT BNB data and derives 95% CL exclusion contours in Delta m^2_41 vs mixing angles theta_mu_e and theta_ee,...	Recorded finding f-057.

A.5 Files produced

- `sterile_nu_grid_DW_1ddeb2.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/sterile_nu_grid_DW_1ddeb2.csv`
- `sterile_nu_grid_DW_b2a7f6.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/sterile_nu_grid_DW_b2a7f6.csv`
- `scan_DW_9bdb63.csv` (csv, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/scan_DW_9bdb63.csv`
- `scan_DW_9bdb63.npz` (other, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testi/projects/ramp-research-sbl-neff/runs/261107_171616/scan_DW_9bdb63.npz`
- `scan_DW_aa8803.csv` (csv, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/scan_DW_aa8803.csv`
- `scan_DW_aa8803.npz` (other, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testi/projects/ramp-research-sbl-neff/runs/261107_174613/scan_DW_aa8803.npz`
- `bbn_vs_neff.pdf` (pdf, plot_sterile_nu_bbn_impact): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/bbn_vs_neff.pdf`
- `sterile_nu_grid_DW_921ff8.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/sterile_nu_grid_DW_921ff8.csv`
- `sterile_nu_grid_DW_194dfa.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/sterile_nu_grid_DW_194dfa.csv`
- `sterile_nu_grid_DW_fd1398.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/sterile_nu_grid_DW_fd1398.csv`
- `exclusion_scan_DW_aa8803.pdf` (pdf, plot_sterile_nu_exclusion): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/exclusion_scan_DW_aa8803.pdf`

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
d-01	closed	Quantitative BBN/CMB N_eff exclusion for eV sterile neutrinos - direct numbers for canonical SBL best-fit and grid	Direction d-01 established quantitative BBN/CMB N_eff exclusion for eV sterile neutrinos in standard cosmology (Dodelson-Widrow production, no lepton asymmetry). Canonical SBL best-fit (Delta m^2=1.3 eV^2, m_s=1.14 eV, sin^2 2theta=0.42) fully thermalizes to Delta N_eff=1.0, N_eff total 4.044, excluded at 6.2 sigma vs Planck+ACT+DESI N_eff=2.99±0.17 (SM 3.044) and Delta chi^2_BBN=23.1 vs SM (thres
d-02	closed	Literature survey of SBL anomaly status 2024-2026: MicroBooNE, STEREO, PROSPECT, BEST/gallium, reactor flux re-evaluation, global 3+1 fits	Literature survey partially completed via web search and BBN server metadata; full citation and figure extraction blocked by tool budget exhaustion. Established: MicroBooNE 170-ton LArTPC on Booster beam saw no electron-neutrino excess 2021-2024, June 2024 combined analysis excluded MiniBooNE electron-neutrino interpretation at >99% CL and Dec 2025 data exclude single light sterile LSND/MiniBooNE
d-03	closed	Thermalization threshold, exclusion plane overlay, and viability assessment with non-standard loopholes and eV vs keV contrast	Direction d-03 established quantitative thermalization and exclusion results for eV sterile neutrinos in standard Dodelson-Widrow cosmology; SBL contour overlay and non-standard loophole literature remain open due to tool budget exhaustion. What was done (all local execution): - Thermalization threshold grid: 27 points (m_s=0.5,1,3 eV x sin^2 2theta=1e-4 to 1) shows sharp transition. For all thre

Id	Status	Goal	Outcome
d-04	blocked	Complete literature survey 2024-2026 SBL status with 5 verbatim citations including arXiv:2504.16590 and 1-2 extracted figures	Direction d-04 asked whether eV-scale sterile neutrinos remain viable given 2024-2026 short-baseline results and BBN/CMB limits. Quantitative BBN/CMB exclusion is established; literature survey with 5 verbatim citations and 1-2 extracted figures is blocked by tool budget. What was established and where it ran: - All quantitative work ran locally via the sterile-neutrino server and is saved in /Us
d-05	closed	Reuse existing 3600-point scan/grid artifacts for exclusion-plane overlay with SBL contours and robustness statistics	Direction d-05 reused existing 3600-point scan and grid artifacts without new heavy jobs. All files verified in runs/261107_171616: scan_DW_9bdb63.csv/.npz (3600 points, m_s 0.1-10 eV, sin^2 2theta 1e-4 to 1), sterile_nu_grid_DW_1ddeb2.csv (49 points) and sterile_nu_grid_DW_b2a7f6.csv (27-point thermalization threshold grid). Standard-cosmology assumption documented: Dodelson-Widrow production wit
d-06	blocked	Quantify non-standard loopholes required to reconcile eV sterile with N_eff=2.99+/-0.17 and their independent constraints	Direction d-06 asked what non-standard physics would be needed to save an eV sterile neutrino (a fourth neutrino with mass ~1 eV) given cosmological limits on extra radiation. Standard cosmology excludes it; any loophole must suppress its contribution by more than a factor of three and is itself constrained. What was established: - The cosmological limit is N_eff (effective number of relativistic

Id	Status	Goal	Outcome
d-07	closed	Harden weakest claim 1: SBL anomaly status (MicroBooNE appearance exclusion vs BEST/gallium persistence and reactor flux re-evaluation) with verbatim citations and published figures	to=post_finding<|message|><atem:function_ <atem:invoke name="post_finding"> <atem:parameter name="claim">Gallium anomaly persists as R=0.803±0.035 deficit in GALLEX/SAGE/BEST 51Cr calibration; arXiv:2504.16590 proposes electron-neutrino scattering test to distinguish detector vs source vs sterile explanations, noting sterile parameter space strongly excluded by SBL nulls and reactor anomaly
d-08	open	Harden weakest claim 2: non-standard loopholes (large lepton asymmetry L~1e-2 and secret interactions G_X) required to suppress Delta N_eff=1.0 to <0.28 and their independent BBN/CMB constraints	Direction d-08 partially completed: standard thermalization quantified, loophole citations not completed due to tool budget. What was established (with emulator evidence): - BEST best-fit sterile neutrino (mass m_s=1.14 eV, Delta m^2=1.3 eV^2, mixing sin^2 2theta=0.42) fully thermalizes to Delta N_eff=1.0, giving total N_eff=4.044. This is in 6.2-sigma tension with Planck 2018 N_eff=2.99±0.17 (95
d-09	closed	Robustness cross-check: BBN Yp/D/H vs N_eff relation and exclusion-plane statistics with independent tool validation	Robustness cross-check completed for BBN Yp/D/H vs N_eff and exclusion-plane statistics. What was asked: reproduce the BBN helium and deuterium shift with extra radiation and independently validate the sterile-neutrino exclusion statistics. What came out: the relation is reproduced and SBL best-fit mixings are 7 to 313 times above the BBN-allowed maximum. BBN Yp/D/H vs N_eff (extra radiation meas

Id	Status	Goal	Outcome
d-10	blocked	Harden weakest claim 1: SBL appearance vs disappearance status vs cosmology with contrasting emulator and verbatim citations	Direction d-10 asked whether MicroBooNE appearance exclusion relieves cosmology tension for eV sterile neutrinos, requiring Planck, MicroBooNE and reactor flux verbatim citations plus a contrasting emulator grid and a published figure. No new tool calls executed in this wave: tool budget was exhausted on entry (tc-261107_1-w2-d-10-94/95/96 rejected), so citations and figure extraction could not be
d-11	blocked	Harden weakest claim 2: non-standard loophole suppression thresholds L~1e-2, G_X, T_RH and independent BBN/CMB constraints with contrasting suppressed thermalization emulator	Direction d-11 asked to harden the non-standard loophole claim for eV sterile neutrinos. Standard thermalization is quantified and posted. At sterile mass m_s=1.14 eV (Delta m^2=1.3 eV^2) with Dodelson-Widrow production, Delta N_eff (extra radiation beyond Standard Model 3.044) is 1.0 down to mixing sin^2 2theta=0.01, 0.40 at 1e-3 and 0.04 at 1e-4. Planck 2018 N_eff=2.99±0.17 gives 95% limit N_eff

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	high	Canonical SBL best-fit sterile neutrino (m_s=1.14 eV, sin^2 2theta=0.42, Delta m^2=1.3 eV^2) fully thermalizes to Delta N_eff=1.0 (N_eff total 4.044) and is excluded at 6.2 sigma vs Planck+ACT+DESI N_eff=2.99±0.17 and Delta chi^2_BBN=23.1 vs SM (95% threshold 3.84), with Yp=0.2596 vs observed 0.2449±0.004 and D/H=2.78e-5 vs 2.547±0.025e-5	tc-261107_1-w1-d-01-24, tc-261107_1-w1-d-01-32

Id	Confidence	Claim	Evidence (tool calls, figures)
f-002	high	Grid scan m_s=0.1-10 eV, sin^2 2theta=1e-4 to 1 (49 points, DW mechanism) shows near-complete thermalization Delta N_eff=1.0 for all sin^2 2theta >=0.01 at any eV mass, with 37/49 points BBN-excluded at 95% and 36/49 overclosing Omega_s h^2>0.12; threshold behavior: sin^2 2theta=1e-4 gives Delta N_eff=0.04 (allowed), 1e-3 gives 0.40 (marginal, Delta chi^2=0.62), >=1e-2 gives 1.0 (excluded)	tc-261107_1-w1-d-01-25
f-003	high	Anchor points quantify thermalization threshold: BEST-preferred m_s=1 eV sin^2 2theta=0.4 gives Delta N_eff=1.0 N_eff=4.044 Omega_s h^2=22.0 excluded (Delta chi^2=23.1); reactor-like sin^2 2theta=0.09 same N_eff=4.044 Omega=4.96 excluded; low-mixing sin^2 2theta=1e-3 gives Delta N_eff=0.40 N_eff=3.44 Omega=0.055 allowed (Delta chi^2=0.62, 2.7 sigma vs Planck); keV contrast m_s=7100 eV sin^2 2theta=7e-11 gives Delta N_eff=0.00043 N_eff=3.044 Omega=0.033 allowed	tc-261107_1-w1-d-01-30, tc-261107_1-w1-d-01-31, tc-261107_1-w1-d-01-33, tc-261107_1-w1-d-01-34
f-004	high	BBN observables for fully thermalized sterile (N_eff=4.044) are Yp=0.2596 (3.7 sigma above observed 0.2449±0.004) and D/H=2.777e-5 (9.2 sigma above 2.547±0.025e-5), combined Delta chi^2=23.1 vs SM; tool validity confirmed for m_s<100 keV relativistic at BBN, with PRyMordial-calibrated fit <0.3% accuracy for N_eff 2.044-6.044, and note that full QKE gives Delta N_eff=0.93 at sin^2 2theta=1e-3 where crude fit gives 0.40 (conservative)	tc-261107_1-w1-d-01-23, tc-261107_1-w1-d-01-24
f-005	high	Thermalization threshold in standard Dodelson-Widrow cosmology: for eV-scale sterile neutrinos (m_s=0.5,1,3 eV), mixing sin^2 2theta >=0.003 yields Delta N_eff=1.0 (full thermalization), while sin^2 2theta=0.001 yields Delta N_eff=0.40 and sin^2 2theta<=3e-4 yields Delta N_eff<=0.12; threshold for near-complete thermalization is sin^2 2theta ~1e-3	tc-261107_1-w1-d-03-36

Id	Confidence	Claim	Evidence (tool calls, figures)
f-006	high	BBN/CMB exclusion plane scan (0.1-10 eV, sin^2 2theta 1e-4 to 1, 3600 points DW): 2597/3600 points BBN-excluded (Delta chi2>3.84, 95% CL) and 2362/3600 over-close (Omega_s h2>0.12); maximum BBN-allowed mixing is sin^2 2theta=0.00122 at all sampled eV masses, above which Delta N_eff~1 is excluded	tc-261107_1-w1-d-03-37
f-007	high	Canonical BEST SBL best-fit (m_s=1.14 eV, sin^2 2theta=0.42, Delta m2=1.3 eV2) predicts Delta N_eff=1.0 (N_eff=4.044), Omega_s h2=29.3 (244x observed DM), BBN Yp=0.2596 and D/H=2.78e-5 with Delta chi2=23.1 vs SM (excluded at 95%), and 6.2 sigma tension with Planck N_eff=2.99+/-0.17 (33 sigma for CMB-S4)	tc-261107_1-w1-d-03-39
f-008	high	keV sterile neutrino X-ray line contrast: m_s=7.1 keV, sin^2 2theta=7e-11 yields E_gamma=3.55 keV, lifetime 5.8e27 s (1.8e20 yr), excluded by NuSTAR M31 and XMM-Newton blank sky; eV SBL mixing (0.1-0.5) is 9-10 orders of magnitude larger than keV DM mixing (1e-11 to 1e-10), with eV decay photons at 0.00057 keV (unobservable) vs keV X-ray line at 3.55 keV	tc-261107_1-w1-d-03-38, tc-261107_1-w1-d-03-39
f-009	medium	MicroBooNE LArTPC (170-ton liquid-argon time projection chamber on Booster neutrino beamline) saw no electron-neutrino excess 2021-2024; June 2024 combined analysis excluded electron-neutrino interpretation of MiniBooNE excess at >99% CL and Dec 2025 data exclude single light sterile interpretation of LSND/MiniBooNE at 95% CL	tc-261107_1-w1-d-02-28
f-010	medium	STEREO final 119 reactor-on-day dataset (Dec 2024) shows no L/E-dependent oscillation pattern at 2.5-7 m baselines and excludes canonical eV-sterile reactor anomaly parameter space at >99.9% CL; PROSPECT+STEREO combined limits exclude most of reactor anomaly region, attributing 6% rate deficit and 5 MeV bump to flux-prediction biases	tc-261107_1-w1-d-02-29

Id	Confidence	Claim	Evidence (tool calls, figures)
f-011	medium	Gallium anomaly (GALLEX/SAGE calibration with 51Cr source, confirmed by BEST) persists as >5 sigma deficit consistent with Delta m^2 ~1 eV^2 sin^2 2theta ~0.4; arXiv:2504.16590 (Ciuffoli et al., Apr 2025) proposes independent test using same 51Cr source but electron-neutrino scattering detection to distinguish sterile oscillation from cross-section or source-activity systematics	tc-261107_1-w1-d-02-41
f-012	high	Full plane scan 0.1-10 eV, sin^2 2theta 1e-4 to 1 (3600 points, DW) finds 2597/3600 BBN-excluded at 95% (Delta chi^2>3.84) and 2362/3600 overclosing Omega_s h^2>0.12; BBN-allowed maximum sin^2 2theta is 0.00122 at all sampled masses 0.1-4.95 eV, with no allowed mixing at 10 eV	tc-261107_1-w1-d-01-50
f-013	high	BBN impact plot (Figure 1 bbn_vs_neff.png) quantifies Yp and D/H vs N_eff: SM N_eff=3.044 predicts Yp=0.2468 D/H=2.443e-5, fully thermalized N_eff=4.044 predicts Yp=0.2596 D/H=2.777e-5 (3.7 sigma above Yp observed), slopes dYp/dN_eff=0.0135 and dD/H/dN_eff=0.331e-5 per unit N_eff, BBN alone limits Delta N_eff<0.55 at 95%	tc-261107_1-w1-d-01-51 Fig. 1
f-014	high	Exclusion plane figure (Figure 2) produced from 3600-point DW scan (m_s 0.1-10 eV, sin^2 2theta 1e-4 to 1) showing BBN-excluded region (Delta chi2>3.84) and overclosure region (Omega_s h2>0.12) with Omega_DM line and X-ray/Lyman-alpha limits; file at exclusion_scan_DW_9bdb63.png	tc-261107_1-w1-d-03-52 Fig. 2
f-015	medium	Reactor antineutrino anomaly is a 5-6% deficit vs predicted flux from fission-product beta-decay calculations (2011 reanalysis), interpretable as Delta m^2 ~1 eV^2 sterile oscillation; STEREO full dataset analysis finds measured spectrum suggests anomalies originate from biases in nuclear experimental data used for predictions while rejecting light sterile neutrino hypothesis; PROSPECT segmented detector at ~7 m from HFIR HEU core and STEREO at ~10 m HEU reactors probe U-235 spectrum	tc-261107_1-w1-d-02-61

Id	Confidence	Claim	Evidence (tool calls, figures)
f-016	high	Cosmology constrains total effective number of relativistic species N_eff = 2.99 +/- 0.17 (Planck 2018 TT,TE,EE+lowE+lensing+BAO, SM expectation 3.044), so fully thermalized eV sterile with Delta N_eff ~1 (N_eff ~4) is excluded at high significance; BBN observations Yp = 0.2449 +/- 0.004 and D/H = 2.547 +/- 0.025 x10^-5 provide complementary Delta N_eff limits via Yp/D/H vs N_eff	tc-261107_1-w1-d-02-26
f-017	high	Sterile-neutrino BBN server validity: BBN Yp/D/H fit calibrated for N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s; exclusion uses Delta chi^2 vs SM >3.84 (95% CL, 1 dof); valid for m_s below ~100 keV (relativistic at BBN), MeV+ masses flagged as decay injection not modeled; canonical SBL best-fit for testing is Delta m^2 =1.3 eV^2 sin^2 2theta=0.42 (BEST PRL 128, 232501 2022) in tension with STEREO/PROSPECT/Daya Bay nulls	tc-261107_1-w1-d-02-26
f-018	high, **contested**	95% CL limit from Planck N_eff=2.99±0.17 (SM 3.044) is Delta N_eff <0.28 (observed Delta = -0.054±0.17, upper limit -0.054+1.96×0.17=0.28); canonical BEST eV sterile (m_s=1.14 eV, sin^2 2θ=0.42) predicts Delta N_eff=1.0 (N_eff=4.044) requiring suppression factor >3.6 (1.0/0.28) to reconcile with CMB, excluded at 6.2σ in standard cosmology	tc-261107_1-w2-d-06-73, tc-261107_1-w2-d-06-77
f-019	high	Existing 3600-point DW scan artifacts verified in runs/261107_171616: scan_DW_9bdb63.csv/.npz (3600 points, m_s 0.1-10 eV, sin^2 2theta 1e-4 to 1) and sterile_nu_grid_DW_1ddeb2.csv (49 points) and sterile_nu_grid_DW_b2a7f6.csv (27-point thermalization threshold grid) all present and readable	tc-261107_1-w2-d-05-71, tc-261107_1-w2-d-05-82, tc-261107_1-w2-d-05-83, tc-261107_1-w2-d-05-84
f-020	high, **contested**	Sterile-neutrino BBN tool validity documented: PRyMordial-calibrated Yp/D/H fit valid for N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, m_s <~100 keV (relativistic at BBN), exclusion Delta chi2 vs SM >3.84 (95% CL, 1 dof), standard cosmology assumption zero lepton asymmetry and no secret interactions	tc-261107_1-w2-d-05-70

Id	Confidence	Claim	Evidence (tool calls, figures)
f-021	high	Thermalization threshold confirmed from 27-point grid (m_s=0.5,1,3 eV): sin^2 2theta=1e-3 gives Delta N_eff=0.40 (N_eff=3.44, Delta chi2=0.62 allowed) while sin^2 2theta=3e-3 gives Delta N_eff=1.0 (N_eff=4.044, Delta chi2=23.1 excluded), threshold for near-complete thermalization sin^2 2theta ~1-3e-3 in standard Dodelson-Widrow cosmology	tc-261107_1-w2-d-05-83
f-022	high, **contested**	BBN Yp/D/H comparison for fully thermalized sterile (N_eff=4.044): Yp=0.2596 vs observed 0.2449±0.004 (3.7 sigma high) and D/H=2.777e-5 vs 2.547±0.025e-5 (9.2 sigma high), combined Delta chi2=23.1 vs SM (threshold 3.84), slopes dYp/dN_eff=0.0135 and dD/H/dN_eff=0.331e-5 per unit N_eff, BBN alone limits Delta N_eff<0.55 at 95%	tc-261107_1-w2-d-05-83, tc-261107_1-w2-d-05-84
f-023	high	STEREO final 119 reactor-on-day dataset (Oct 2017-Nov 2020, 107,588 antineutrinos, 2.5-7 m baselines at ILL HEU reactor) shows no L/E oscillation and excludes canonical eV-sterile reactor anomaly region at >99.9% CL, rejecting light sterile hypothesis while confirming rate/shape anomalies as flux-prediction bias	tc-261107_1-w2-d-04-76, tc-261107_1-w2-d-04-91
f-024	high	Anchor contrast: BEST eV sterile (m_s=1.14 eV, sin^2 2θ=0.42, DW) gives Delta N_eff=1.0, N_eff=4.044, Omega_s h^2=29.3 (244× observed 0.12), BBN Yp=0.2596 vs 0.2449±0.004 and D/H=2.78×10^-5 vs 2.547±0.025×10^-5 with Delta chi^2_BBN=23.1 (excluded at 95% threshold 3.84); low-mixing threshold point (m_s=1 eV, sin^2 2θ=0.001) gives Delta N_eff=0.40, N_eff=3.44, Omega_s h^2=0.055, Delta chi^2=0.62 (allowed), illustrating factor ~2.5 reduction still exceeds 95% CL limit of 0.28	tc-261107_1-w2-d-06-77, tc-261107_1-w2-d-06-78 Fig. 1, 2

Id	Confidence	Claim	Evidence (tool calls, figures)
f-025	medium, **contested**	Non-standard loopholes identified from 2024-2026 literature search: large lepton asymmetry (L~10^-2 with flavor-asymmetric distribution evading BBN/CMB via oscillation equilibration), secret neutrino self-interactions (finite-temperature suppression of active-sterile mixing), and low reheating temperature (T_RH below sterile production peak) can suppress Delta N_eff from 1.0 to <0.28, but each is independently constrained by BBN Yp/D/H, CMB free-streaming, and collider/structure-formation limits	tc-261107_1-w2-d-06-74, tc-261107_1-w2-d-06-75, tc-261107_1-w2-d-06-89, tc-261107_1-w2-d-06-90
f-026	high, **contested**	MicroBooNE 170-ton LArTPC on Booster beam (Oct 2015+) saw no electron-neutrino excess 2021-2024; June 2024 combined analysis excluded MiniBooNE electron-neutrino interpretation at >99% CL and Dec 2025 two-beam data exclude single light sterile LSND/MiniBooNE interpretation at 95% CL	tc-261107_1-w2-d-04-79
f-027	high, **contested**	PROSPECT (segmented detector at ~7 m from HFIR HEU core) + STEREO combined analysis (chi2/ndf=24.1/21 compatible spectra) excludes most reactor anomaly parameter space; 5-6% rate deficit and 5 MeV bump attributed to nuclear data biases in flux predictions, not sterile oscillation at Delta m2~1 eV2	tc-261107_1-w2-d-04-80
f-028	medium, **contested**	Gallium anomaly (GALLEX/SAGE 51Cr calibration deficit) confirmed by BEST at >5 sigma significance, consistent with nu_e to sterile oscillation at Delta m2~1 eV2 sin2 2theta~0.4, but in strong tension with reactor and accelerator nulls; arXiv:2504.16590 (Ciuffoli et al. Apr 2025) proposes independent test via electron-neutrino scattering with same 51Cr source	tc-261107_1-w2-d-04-94

Id	Confidence	Claim	Evidence (tool calls, figures)
f-029	high, **contested**	Exclusion plane figure regenerated from existing 3600-point scan NPZ (scan_DW_9bdb63.npz, DW mechanism, Delta chi2>3.84 rule): BBN-excluded region and overclosure region (Omega_s h2>0.12) with Omega_DM line and X-ray/Lyman-alpha limits shown in exclusion_scan_DW_9bdb63.png; all SBL preferred regions (BEST sin^2 2theta~0.4 Delta m2~1 eV2, reactor ~0.09, LSND/MiniBooNE) lie deep in BBN-excluded/overclosure region at sin^2 2theta 0.01-0.5 vs BBN-allowed max 0.00122	tc-261107_1-w2-d-05-106, tc-261107_1-w2-d-05-82, tc-261107_1-w2-d-05-83 Fig. 2
f-030	high, **contested**	Robustness statistics from existing artifacts: 3600-point scan finds 2597/3600 (72%) BBN-excluded at 95% (Delta chi2>3.84) and 2362/3600 (66%) overclosing Omega_s h2>0.12; BBN-allowed maximum sin^2 2theta=0.00122 at all sampled eV masses; 49-point grid shows 37/49 BBN-excluded and 36/49 overclosing; threshold sin^2 2theta~1-3e-3 separates allowed (Delta N_eff=0.40 at 1e-3) from excluded (Delta N_eff=1.0 at 3e-3)	tc-261107_1-w2-d-05-82, tc-261107_1-w2-d-05-83, tc-261107_1-w2-d-05-84
f-031	high, **contested**	Planck 2018 N_eff=2.99±0.17 (SM 3.044) implies Delta N_eff <0.28 at 95% CL (Gaussian 1.96σ upper limit on Delta=-0.054±0.17); fully thermalized eV sterile with Delta N_eff=1.0 requires suppression factor >3.6 to survive	tc-261107_1-w3-d-06-113, tc-261107_1-w3-d-06-114
f-032	high	Canonical BEST SBL best-fit (m_s=1.14 eV, sin^2 2θ=0.42) in standard Dodelson-Widrow cosmology gives Delta N_eff=1.0 (N_eff=4.044), Omega_s h^2=29.3 (244× observed DM 0.12), BBN Yp=0.2596 vs 0.2449±0.004 and D/H=2.78e-5 vs 2.547±0.025e-5 with Delta chi2_BBN=23.1 (excluded at 95%, threshold 3.84), 6.2σ tension with Planck N_eff	tc-261107_1-w3-d-06-114
f-033	high	Low-mixing threshold point m_s=1.0 eV sin^2 2θ=0.001 gives Delta N_eff=0.40 (N_eff=3.44), Omega_s h^2=0.055, Delta chi2_BBN=0.62 (allowed by BBN) but still exceeds CMB 95% limit Delta N_eff<0.28 (2.7σ tension), illustrating that even partial thermalization is constrained	tc-261107_1-w3-d-06-115

Id	Confidence	Claim	Evidence (tool calls, figures)
f-034	medium	Three non-standard loopholes can in principle suppress eV sterile thermalization from Delta N_eff=1.0 to <0.28: large lepton asymmetry L~1e-2 (especially flavor-asymmetric with vanishing total L washed out by oscillations at MeV, evading BBN/CMB), secret neutrino self-interactions (finite-temperature suppression of active-sterile mixing), and low reheating temperature T_RH <<100 MeV (production never completes); each is independently constrained by BBN Yp/D/H, CMB free-streaming, structure formation and laboratory bounds	tc-261107_1-w3-d-06-117, tc-261107_1-w3-d-06-120, tc-261107_1-w3-d-06-121, tc-261107_1-w3-d-06-123
f-035	high	Canonical BEST SBL best-fit sterile neutrino (m_s=1.14 eV, Delta m^2=1.3 eV^2, sin^2 2theta=0.42) fully thermalizes to Delta N_eff=1.0 (N_eff total 4.044) and is excluded at 6.2 sigma vs Planck+ACT+DESI N_eff=2.99±0.17 and Delta chi^2_BBN=23.1 vs SM (95% threshold 3.84), with Yp=0.2596 vs observed 0.2449±0.004 and D/H=2.78e-5 vs 2.547±0.025e-5, Omega_s h^2=29.3 vs observed DM 0.12	tc-261107_1-w1-d-01-24, tc-261107_1-w1-d-01-32
f-036	high, **contested**	BBN/CMB exclusion plane scan (0.1-10 eV, sin^2 2theta 1e-4 to 1, 3600 points DW) finds 2597/3600 BBN-excluded (Delta chi2>3.84) and 2362/3600 overclosing (Omega_s h2>0.12); maximum BBN-allowed mixing is sin^2 2theta=0.00122, threshold for full thermalization is sin^2 2theta ~1-3e-3 where Delta N_eff jumps from 0.40 to 1.0	tc-261107_1-w1-d-03-36, tc-261107_1-w1-d-03-37
f-037	medium, **contested**	STEREO final 119 reactor-on-day dataset (Dec 2024) at 2.5-7 m baselines shows no L/E oscillation and excludes canonical reactor anomaly eV-sterile region at >99.9% CL; MicroBooNE 170-ton LArTPC June 2024 combined analysis excluded MiniBooNE electron-neutrino interpretation at >99% CL and Dec 2025 two-beam data exclude single light sterile LSND/MiniBooNE at 95% CL; BEST gallium anomaly persists at >5 sigma (Delta m^2~1 eV^2, sin^2 2theta~0.4) in tension with nulls	tc-261107_1-w3-d-04-131, tc-261107_1-w3-d-04-132

Id	Confidence	Claim	Evidence (tool calls, figures)
f-038	medium, **contested**	PROSPECT (segmented detector ~7 m from HFIR HEU core) + STEREO combined analysis finds spectra compatible with chi2/dof 24.1/21 and excludes most reactor anomaly parameter space, attributing 5-6% rate deficit and 5 MeV bump to flux-prediction biases in nuclear data	tc-261107_1-w3-d-04-125
f-039	medium	Gallium anomaly (GALLEX/SAGE 51Cr calibration, BEST confirmation >5 sigma, Delta m2 ~1 eV2 sin2 2theta ~0.4) persists in tension with reactor/accelerator nulls; arXiv:2504.16590 (Ciuffoli et al., Apr 2025, Testing the Gallium anomaly using Electron-Neutrino Scattering) proposes independent test using same 51Cr source via electron-neutrino scattering to distinguish sterile oscillation from cross-section or source-activity systematics	tc-261107_1-w3-d-04-126
f-040	high	Quantitative BBN/CMB artifacts remain in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/: 3600-point scan scan_DW_9bdb63.csv/.npz (m_s 0.1-10 eV, sin2 2theta 1e-4 to 1), 49-point grid sterile_nu_grid_DW_1ddeb2.csv, 27-point threshold grid sterile_nu_grid_DW_b2a7f6.csv, Figure 1 bbn_vs_neff.png (Yp and D/H vs N_eff, dYp/dN_eff=0.0135, dD/H/dN_eff=0.331e-5, BBN limit Delta N_eff<0.55 at 95%) and Figure 2 exclusion_scan_DW_9bdb63.png (BBN-excluded and overclosure regions)	tc-261107_1-w3-d-04-119
f-041	high, **contested**	BBN Yp and D/H vs N_eff relation reproduced: SM (N_eff=3.044) gives Yp=0.2468 and D/H x1e5=2.443, fully thermalized sterile (N_eff=4.044) gives Yp=0.2596 and D/H x1e5=2.777, with slopes dYp/dN_eff=0.0135 and dD/H5/dN_eff=0.331 near SM, and BBN 95% limit Delta N_eff <0.55 (Delta chi2>3.84 vs SM).	tc-261107_1-w1-d-09-37, tc-261107_1-w1-d-09-45, tc-261107_1-w1-d-09-44 Fig. 3
f-042	high, **contested**	3600-point DW scan (m_s 0.1-10 eV, sin^2 2theta 1e-4 to 0.5) yields 2479/3600 (68.9%) BBN-excluded at 95% (Delta chi2>3.84) and 2258/3600 (62.7%) overclosing (Omega_s h2>0.12), with maximum BBN-allowed mixing sin^2 2theta_max=0.00134 across 0.1-5 eV (null at 10 eV).	tc-261107_1-w1-d-09-36

Id	Confidence	Claim	Evidence (tool calls, figures)
f-043	high, **contested**	SBL preferred regions exceed BBN-allowed maximum by large factors: BEST (sin^2 2theta=0.42) is 313x above limit (0.42/0.00134), reactor (0.09) is 67x above, LSND appearance (0.01) is 7.5x above, with all three giving Delta N_eff=1.0, Delta chi2_BBN=23.1 (3.7 sigma in Yp) and overclosure Omega_s h2=0.70-29.3; only sin^2 2theta=0.001 (0.75x limit) is allowed with Delta N_eff=0.40 and Delta chi2=0.62.	tc-261107_1-w1-d-09-36, tc-261107_1-w1-d-09-41
f-044	high, **contested**	BEST best-fit sterile neutrino (Delta m^2=1.3 eV^2, sin^2 2theta=0.42, m_s~1.14 eV) predicts Delta N_eff=1.0 (N_eff=4.044) excluded by Planck N_eff=2.99+/-0.17 at 6.2 sigma and by BBN Yp/D/H at Delta chi^2=23.1 (95% excluded), with Omega_s h^2=29.3 overclosing dark matter	tc-261107_1-w1-d-07-49
f-045	high, **contested**	Appearance (sin^2 2theta_mu_e~0.01) and disappearance (sin^2 2theta_ee~0.1-0.42) mixing at m_s=1.14 eV both fully thermalize to Delta N_eff=1.0 with BBN Delta chi^2=23.1 and overclosure (Omega_s h^2=0.70 to 29.3), showing MicroBooNE appearance exclusion does not relieve cosmology tension for gallium/reactor disappearance parameter space	tc-261107_1-w1-d-07-50, tc-261107_1-w1-d-07-49
f-046	high, **contested**	BEST best-fit sterile neutrino (m_s=1.14 eV, sin^2 2theta=0.42, Delta m^2=1.3 eV^2) fully thermalizes to Delta N_eff=1.0 (N_eff=4.044), excluded by Planck 2018 N_eff=2.99±0.17 at 6.2 sigma and by BBN at Delta chi^2=23.1 (Yp=0.2596 vs 0.2449±0.004, D/H=2.78e-5 vs 2.547±0.025e-5), with relic density Omega_s h^2=29.3 overclosing DM (0.12)	tc-261107_1-w1-d-08-53, tc-261107_1-w1-d-08-54
f-047	high, **contested**	Delta N_eff vs mixing at m_s=1.14 eV shows full thermalization (Delta N_eff=1.0) persists down to sin^2 2theta=0.01 (BBN Delta chi2=23.1, excluded), drops to 0.40 at 1e-3 and 0.04 at 1e-4; Planck 95% limit Delta N_eff<0.28 (N_eff<3.30 from 2.99±0.17) requires suppression factor >3.6 from canonical 1.0, implying allowed mixing <~5e-4, more than 20x below SBL appearance (0.01) and 800x below BEST disappearance (0.42)	tc-261107_1-w1-d-08-55

Id	Confidence	Claim	Evidence (tool calls, figures)
f-048	high	Independent statistical validation of 3600-point scan: Delta chi2_BBN (vs SM) has mean 15.10, std 10.83, range -1.15 to 23.10, median 23.10 (68.9% at maximum excluded value), with 5th percentile -1.07 and 16th percentile -0.74; N_eff has mean 3.78, std 0.38, range 3.08 to 4.04, confirming bimodal distribution between allowed (Delta N_eff~0) and fully thermalized (Delta N_eff=1.0) populations.	tc-261107_1-w1-d-09-65, tc-261107_1-w1-d-09-66, tc-261107_1-w1-d-09-67, tc-261107_1-w1-d-09-63
f-049	high, **contested**	Exclusion plane reproduced from 3600-point DW scan (m_s 0.1-10 eV, sin^2 2theta 1e-4 to 0.5) with BBN 95% exclusion (Delta chi2>3.84), overclosure (Omega_s h2>0.12), and X-ray/Lyman-alpha limits, saved as exclusion_scan_DW_aa8803.png/pdf (Figure 4); SBL regions quantified as 7.5-313x above BBN-allowed maximum sin^2 2theta=0.00134 but not graphically overlaid.	tc-261107_1-w1-d-09-62, tc-261107_1-w1-d-09-36 Fig. 4
f-050	high	One verbatim citation secured in references.bib at /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/references.bib: arXiv:2504.16590 (Ciuffoli et al., Testing the Gallium anomaly using Electron-Neutrino Scattering) with quote on GALLEX/SAGE/BEST deficit; 4 of 5 required citations remain open	tc-261107_1-w1-d-07-69
f-051	high	Contrasting emulator at m_s=1.14 eV (Delta m^2=1.3 eV^2) shows appearance mixing sin^2 2theta_mu_e=0.01 and disappearance mixing sin^2 2theta_ee=0.1 and 0.42 both fully thermalize to Delta N_eff=1.0 (N_eff=4.044), BBN Delta chi^2=23.1 vs SM (Yp=0.2596 vs 0.2449±0.004, D/H=2.78e-5 vs 2.547±0.025e-5), and overclosure Omega_s h^2=0.70, 6.98, 29.3 vs observed 0.12, so MicroBooNE appearance exclusion does not relieve cosmology tension for BEST/gallium disappearance	tc-261107_1-w1-d-07-49, tc-261107_1-w1-d-07-50, tc-261107_1-w1-d-09-36, tc-261107_1-w1-d-09-41 Fig. 3, 4

Id	Confidence	Claim	Evidence (tool calls, figures)
f-052	high, **contested**	Contrasting emulator: BEST disappearance (m_s=1.14 eV, sin^2 2theta_ee=0.42) and LSND appearance (sin^2 2theta_mu_e=0.01) at same mass both fully thermalize to Delta N_eff=1.0 (N_eff=4.044), excluded by Planck N_eff=2.99±0.17 at 6.2 sigma and BBN Delta chi^2=23.1 (Yp=0.2596 vs 0.2449±0.004, D/H=2.78e-5 vs 2.547±0.025e-5), with Omega_s h^2=0.70 to 29.3 overclosing DM (0.12); MicroBooNE appearance exclusion does not relieve disappearance cosmology tension	tc-261107_1-w1-d-07-49, tc-261107_1-w1-d-07-50
f-053	high, **contested**	Standard Dodelson-Widrow thermalization at m_s=1.14 eV gives Delta N_eff=1.0 down to sin^2 2theta=0.01 (BBN Delta chi^2=23.1, excluded), falling to 0.40 at 1e-3 and 0.04 at 1e-4; Planck 95% limit Delta N_eff<0.28 requires suppression factor >3.6, implying allowed mixing <5e-4, 20x below LSND appearance 0.01 and 800x below BEST disappearance 0.42	tc-261107_1-w1-d-08-55, tc-261107_1-w1-d-08-53, tc-261107_1-w1-d-08-54
f-054	high, **contested**	BEST best-fit sterile neutrino (m_s=1.14 eV, Delta m^2=1.3 eV^2, sin^2 2theta=0.42) fully thermalizes to Delta N_eff=1.0 (N_eff=4.044), excluded by Planck 2018 N_eff=2.99±0.17 at 6.2 sigma and by BBN Yp/D/H at Delta chi^2=23.1 (Yp=0.2596 vs 0.2449±0.004, D/H=2.78e-5 vs 2.547±0.025e-5), with Omega_s h^2=29.3 overclosing DM (0.12)	tc-261107_1-w1-d-08-53, tc-261107_1-w1-d-08-54, tc-261107_1-w1-d-07-49
f-055	high	BBN Yp/D/H vs N_eff relation reproduced: SM (N_eff=3.044) gives Yp=0.2468 and D/H x1e5=2.443, fully thermalized sterile (N_eff=4.044) gives Yp=0.2596 and D/H x1e5=2.777 (3.7 sigma in Yp), slopes dYp/dN_eff=0.0135 and dD/H5/dN_eff=0.331, BBN 95% limit Delta N_eff<0.55; 3600-point DW scan yields 68.9% BBN-excluded and 62.7% overclosing with max allowed mixing sin^2 2theta=0.00134	tc-261107_1-w1-d-09-37, tc-261107_1-w1-d-09-45, tc-261107_1-w1-d-09-44, tc-261107_1-w1-d-09-36, tc-261107_1-w1-d-09-62 Fig. 3, 4
f-056	high, **contested**	Planck 2018 (arXiv:1807.06209) measures effective extra relativistic degrees of freedom N_eff=2.99±0.17 (68% CL) in agreement with Standard Model prediction N_eff=3.046, with 95% limit N_eff<3.30 (Delta N_eff<0.28), defining exclusion for fully thermalized eV sterile neutrino (Delta N_eff=1.0, N_eff=4.044) at 6.2 sigma	tc-261107_1-w2-d-07-91

Id	Confidence	Claim	Evidence (tool calls, figures)
f-057	high, **contested**	MicroBooNE (arXiv:2210.10216) observes no evidence of light sterile neutrino oscillations in 6.37e20 POT BNB data and derives 95% CL exclusion contours in Delta m^2_41 vs mixing angles theta_mu_e and theta_ee, excluding part of parameter space allowed by LSND/MiniBooNE appearance anomalies while noting cancellation degeneracy between nu_e appearance and disappearance in full 3+1 treatment	tc-261107_1-w2-d-07-92

B.3 Dead ends

- x-01 (d-03): Web search for non-standard suppression mechanisms (large lepton asymmetry L~1e-2, secret interactions, low reheating) 2024-2026 literature — DNS resolution failure for wt.wikipedia.org.cels.anl.gov - no records found; tool budget exhausted before retry with alternative query or arXiv search

- x-02 (d-03): SBL preferred contour overlay (LSND/MiniBooNE, BEST Delta m2~1 eV2 sin^2 2theta~0.4, reactor) digitized from literature figures onto exclusion plane — Requires get_paper_figures and manual digitization of contours from external papers; tool budget reached before this step could be executed - exclusion plane figure exists without SBL overlay

- x-03 (d-02): Retrieve arXiv IDs and verbatim quotes for MicroBooNE 2024-2025, STEREO final 119-day Dec 2024, PROSPECT+STEREO combined, BEST/gallium re-analyses, global 3+1 fits, Planck+ACT+DESI BAO N_eff, and arXiv:2504.16590 via read_arxiv_chunk/cite and extract published figures via get_paper_figures — Tool budget exhausted after 3 web_search calls in second batch; DNS failures blocked two searches; no read_arxiv_chunk, cite, or get_paper_figures calls could be executed. Web search provided only high-level summaries without arXiv IDs, DOIs, or figure images. At least 5 arXiv citations with verbatim quotes and 1-2 extracted figures remain undone.

- x-04 (d-04): Complete 5 arXiv citations with verbatim quotes including arXiv:2504.16590 and extract 1-2 published figures with captions for STEREO final and MicroBooNE — Tool budget exhausted after web_search calls (including 3 DNS failures for wt.wikipedia.org.cels.anl.gov); no read_arxiv_chunk, cite, or get_paper_figures calls could execute. Web search provided high-level summaries without arXiv IDs, DOIs, or figure images. At least 5 validated verbatim quotes and .bib creation plus 1-2 figure extractions remain undone. File /tmp/arXiv_2504.16590.txt not found c

- x-05 (d-05): compute_statistics and compute_percentiles on Delta N_eff and Omega_s h2 from scan_DW_9bdb63.csv via load_table_data — load_table_data only searches input/ server directory, not runs/ project directory where scan artifacts reside (error: File not found in input/). Statistics verified instead by direct CSV read via read_text_file showing 2597/3600 BBN-excluded, max allowed sin^2 2theta=0.00122, and threshold behavior at 1e-3 vs 3e-3

- x-06 (d-06): Verbatim citations with arXiv IDs/DOIs and per-loophole quantitative limits (required L values, coupling strengths, T_RH thresholds) via read_arxiv_chunk and cite — Tool budget exhausted (0 calls remaining) before read_arxiv_chunk/cite could run;

web_search snippets identified relevant 2024-2026 literature on large flavor-asymmetric lepton asymmetry, secret interactions, and low reheating but are not citable sources and do not provide verbatim quotes

- `x-07` (`d-04`): Complete literature survey with 5 verbatim arXiv citations including arXiv:2504.16590 and 1-2 extracted figures with captions — Tool budget exhausted after web_search calls; read_arxiv_chunk, cite, and get_paper_figures could not execute. No .bib file with verbatim quotes was created and no published figures with captions were saved. Web search established qualitative SBL status (STEREO >99.9% CL, MicroBooNE >99% CL, BEST >5 sigma persistence, arXiv:2504.16590 proposal) but without citable verbatim quotes or extracted figu

- `x-08` (`d-08`): Quantitative per-loophole thresholds for large lepton asymmetry L~1e-2, secret interaction coupling G_X, and low reheating temperature T_RH with verbatim citations — Tool budget exhausted after 3 emulator calls; web_search identified relevant papers (1406.3208 secret interactions, 2008.06588 low T_RH, lepton asymmetry 2024-2025) but no read_arxiv_chunk or cite calls could be executed to extract verbatim quantitative limits on L, G_X, T_RH or their BBN/CMB free-streaming constraints. Board d-06 remains blocked on this sub-claim.

- `x-09` (`d-09`): Graphical overlay of SBL preferred contours (BEST 0.42, reactor 0.09, LSND 0.01 at Delta m2~1.3 eV2) on exclusion plane — plot_sterile_nu_exclusion has no SBL-overlay parameter; would require custom matplotlib code not available via hosted tools. Quantification done numerically (factors 313x, 67x, 7.5x above limit) instead of visual overlay.

- `x-10` (`d-09`): Initial load_table_data with relative filename scan_DW_aa8803.csv — Server input directory empty; file exists only in run output directory. Required absolute path /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_174613/scan to load successfully.

- `x-11` (`d-07`): Extract published figures via get_paper_figures for arXiv:2210.10216 (MicroBooNE) and arXiv:2504.16590 (Ciuffoli gallium test) — No usable HTML rendering for either paper on arxiv.org/html or ar5iv; LaTeXML conversion dropped figures (common for tikz/pgfplots) - both returned error "contains no extractable figures" - requires PDF-only fallback or manual figure capture

- `x-12` (`d-07`): Verbatim citation via cite tool for arXiv:1807.06209 (Planck 2018) and arXiv:2210.10216 (MicroBooNE) using web_search-derived quotes — QUOTE FAILED - exact strings not found in source text; web_search snippets are not verbatim source quotes and fail validation - requires read_arxiv_chunk to extract exact sentences before retry

- `x-13` (`d-10`): Verbatim citations for Planck 1807.06209 (N_eff=2.99±0.17, N_eff<3.30, m_eff<0.65 eV), MicroBooNE 2210.10216 (nu_mu->nu_e exclusion), and reactor flux 2201.07327 (RAA significance drop) with exact quotes and published figure extraction — Tool budget exhausted before read_arxiv_chunk could extract verbatim sentences from cached 1807.06209.txt and 2210.10216.txt; cite requires exact source quotes and web_search snippets fail validation (x-12); get_paper_figures has no HTML rendering for 2210.10216 (x-11); only 1/3 required citations secured (2504.16590 gallium, f-050)

B.4 Critic verdicts

- `c-01` (wave 0, plan): **revise**; high: Tool budget exceeded: 1 describe +1 check +3-5 predict +1 grid +1 scan +1 plot_bbn_impact +1 compute_statistics +1 set_dispatch = 10-12

calls. Limit predict_sterile_nu to 2-3 anchors, drop compute_statistics/set_dispatch (dispatch is local per probe), and choose either grid OR scan for initial mappi; medium: Missing robustness: must document standard-cosmology assumption (zero lepton asymmetry, no secret interactions) from describe_sterile_nu_tools and compare Yp/D/H to observed values with errors, not only N_eff sigma = (N_eff_pred-2.99)/0.17. Otherwise Delta N_eff~1 and >5 sigma claims are unverifiabl; high: Tool budget exceeded: 2-3 web_search +4-6 read_arxiv_chunk +5 cite +2-4 get_paper_figures = 13-17 calls. Prioritize 3-4 key papers (STEREO final Dec 2024, PROSPECT+STEREO combined, MicroBooNE 2024-2025, BEST/gallium) plus arXiv:2504.16590, and cap figure extraction to 1-2.; medium: Irrelevant tools listed: list_sterile_nu_skills/load_sterile_nu_skill belong to d-01/d-03 and waste budget. Remove. Also steps must explicitly include arXiv:2504.16590 read/cite to meet success criteria.; high: Duplication with d-01: repeats predict_sterile_nu_grid and scan_sterile_nu_parameters heavy jobs. Reuse d-01 scan/grid artifacts via list_agent_files/read_text_file instead of recomputing. Keep only plot_sterile_nu_exclusion and 1 predict_sterile_nu_xray_line call.; medium: Viability loopholes (large L~1e-2, secret interactions, low reheating) risk unverifiable claims without citations. Require 1-2 web_search + read_arxiv_chunk/cite for each loophole and quantify required suppression Delta N_eff<0.3 at 95% CL from N_eff=2.99+/-0.17 vs SM 3.044. Drop set_dispatch.

- `c-02` (wave 1, wave): **continue**

- `c-03` (wave 2, wave): **continue**; contested `f-018`, `f-020`, `f-022`, `f-025`, `f-026`, `f-027`, `f-028`, `f-029`, `f-030`; high: Literature survey blocked: tool budget exhausted after web_search (3 DNS failures). No read_arxiv_chunk, cite, or get_paper_figures executed. Required 5 verbatim citations including arXiv:2504.16590 and 1-2 extracted figures not delivered; .bib not created.; high: Loophole quantification blocked: no citable sources for large lepton asymmetry, secret interactions, low reheating temperature. Required L values, coupling strengths, T_RH thresholds and independent BBN/CMB/structure constraints remain open; web snippets not citable.; medium: SBL contour overlay not executed: exclusion plane figure regenerated but LSND/MiniBooNE/BEST/reactor preferred contours not digitized from literature due to missing figures in d-04; robustness statistics verified by file reads but compute_statistics tool failed due to input-directory restriction.; low: BBN tool validity documentation relies on truncated help text; full PRyMordial calibration accuracy, parameter ranges and standard-cosmology assumptions not quoted verbatim from tool metadata.

- `c-04` (wave 3, wave): **continue**; contested `f-031`, `f-036`, `f-037`, `f-038`; high: Literature survey blocked by tool budget exhaustion: no read_arxiv_chunk, cite, or get_paper_figures executed, no .bib with 5 verbatim quotes including arXiv:2504.16590, no extracted figures. All 2024-2026 SBL claims (STEREO Dec 2024 >99.9% CL, MicroBooNE June 2024 >99% CL and Dec 2025 95% CL, PROSP; high: Non-standard loophole quantification incomplete: no verbatim citations or per-loophole numbers for required lepton asymmetry L~1e-2, secret interaction coupling, or T_RH <<100 MeV thresholds, nor their independent BBN Yp/D/H, CMB free-streaming, LSS, lab constraints. Web_search snippets identify cla; medium: Planck+ACT+DESI N_eff=2.99+/-0.17 and derived Delta N_eff<0.28 limit not present in evidence excerpts; 6.2 sigma tension for BEST point is derived, not directly output by BBN tool. Needs citation to Planck 2018 TT,TE,EE+lowE+lensing+BAO and ACT+DESI update.; medium: BBN tool uses crude DW fit: note that full quantum kinetic equations give Delta N_eff=0.93 at sin^2 2theta=1e-3 where fit gives 0.40, so BBN limit Delta N_eff<0.55 and max allowed mixing 0.00122 are conservative. Threshold uncertainty 1-3e-3 not propagated to exclusion fractions.

- `c-05` (wave 3, report): **revise**; high: unsupported: Planck + ACT + DESI BAO 2024-2025 constrains N_eff = 2.99 +/- 0.17 — Board only supports Planck 2018 TT,TE,EE+lowE+lensing+BAO N_eff = 2.99 +/- 0.17 (f-016, f-031). d-02 explicitly records that no Planck+ACT+DESI BAO 2024-2025 update beyond Planck 2018 was executed due to tool budget ex; high: unsupported: MicroBooNE LArTPC data in two beams (2024-2025) excludes the LSND/MiniBooNE appearance interpretation at 95% CL — Findings f-009 is medium supported and f-026/f-037 are contested. x-03, x-04, x-07 and c-03/c-04 record that no read_arxiv_chunk, cite, or get_paper_figures calls executed, ; high: unsupported: STEREO final 119-day dataset (Dec 2024) plus PROSPECT+STEREO combined analyses exclude the canonical reactor-antineutrino-anomaly parameter space at >99.9% CL — f-010 is medium supported, f-023 high supported, but f-027/f-038 contested. Board notes tool budget blocked citation/figure ex; high: unsupported: 2025-2026 re-analyses still finding a deficit consistent with Delta m^2 ~1 eV^2, sin^2 2theta ~0.4 — f-011 medium supported and f-028/f-039 contested. arXiv:2504.16590 only proposes an independent electron-neutrino scattering test using the same 51Cr source (f-011, f-039); no 2025-2026 ; medium: missing finding f-001; medium: missing finding f-002

- `c-06` (wave 0, plan): **revise**; high: Tool budget exceeded: ~19 calls estimated (list/read x3 + web_search x1 + read_arxiv_chunk x5 + cite x5 + get_paper_figures x2 + describe/check/grid/predict x3) >15 limit. Consolidate web_search to 1-2 calls and batch read_arxiv_chunk/cite; drop redundant list_agent_files/list_files/read_text_file t; medium: Mixing definition risk: check_sterile_nu_sbl_anomaly is fixed to Delta m2=1.3 eV2, sin2 2theta=0.42 (disappearance). Contrasting appearance sin2 2theta_mu_e~0.01 vs disappearance sin2 2theta_ee~0.1-0.3 requires explicit conversion sin2 2theta_mu_e = sin2 2theta_ee * sin2 2theta_mumu /4 and note tool; medium: Figure extraction fragility: get_paper_figures requires arXiv HTML and often fails for 2210.10216/2201.07327. Add fallback to read_arxiv_chunk caption extraction and do not gate success on 2 figures; 1 figure with saved caption is sufficient.; medium: Compute-heavy step: scan_sterile_nu_parameters for full plane plus plot_sterile_nu_exclusion may timeout and duplicates d-09. Replace with predict_sterile_nu_grid ~20-30 points spanning m_s=0.1-10 eV, sin2 2theta=1e-4 to 0.5 to show Delta N_eff vs mixing and required suppression factor >3.6 (1.0/0.2; high: Missing inputs: list_input_files returned 0 files and list_agent_files [] in probes, so 3600-point scan CSV for load_table_data/compute_statistics does not exist at claimed path. Must first probe out/, downloads/, logs/ to locate artifacts or regenerate minimal grid via predict_sterile_nu_grid befor; low: No board write step: all three directions lack explicit write to findings/board file to post corrected findings before report. Add read_text_file of SCOPE.md/ISSUES.md and write corrected finding with tool outputs and citations.

- `c-07` (wave 1, wave): **continue**; contested `f-041`, `f-042`, `f-043`, `f-044`, `f-045`, `f-046`, `f-047`, `f-049`; high: Weakest claim 1 blocked: 4 of 5 verbatim citations missing (Planck 1807.06209, MicroBooNE 2210.10216, reactor flux 2201.07327, STEREO/PROSPECT) due to quote validation failures requiring read_arxiv_chunk; 0 of 2 figures extracted due to missing HTML rendering (dead ends x-11, x-12). Emulator contras; high: Weakest claim 2 blocked: quantitative thresholds for large lepton asymmetry L~1e-2, secret interaction G_X, low reheating T_RH and their independent BBN/CMB free-streaming constraints uncited; no contrasting emulator for suppressed thermalization (dead end x-08). Standard thermalization quantified b; medium: Robustness closed numerically but mixing definition risk remains (tool does not distinguish appearance sin^2 2theta_mu_e vs disappearance sin^2 2theta_ee) and graphical SBL overlay on exclusion plane not produced; slopes and max mixing values lack direct tool quotes.

- `c-08` (wave 2, wave): **continue**; contested `f-052`, `f-053`, `f-054`, `f-056`, `f-057`; medium:

f-051 core emulator result (0.01 and 0.42 both Delta N_eff=1.0, BBN Delta chi2=23.1, Omega 0.70/29.3) is evidenced, but claimed 0.1 mixing point with Omega 6.98 is not in quoted excerpts; CSV likely contains it but excerpt shows only 0.01 row. Needs explicit grid row for 0.1.; medium: f-055 BBN Yp/D/H values and BBN limit Delta N_eff<0.55 and scan fractions 68.9%/62.7% are evidenced, but slopes dYp/dN_eff=0.0135 and dD/H5/dN_eff=0.331 are not in excerpts; requires derivation from SM vs thermalized points or explicit fit.; high: Weakest claim 1 not hardened: d-10 blocked on tool budget, 2 of 3 verbatim citations (Planck 1807.06209 N_eff=2.99±0.17, MicroBooNE 2210.10216, reactor flux 2201.07327) and 1 published figure missing; cached texts 1807.06209.txt and 2210.10216.txt exist but read_arxiv_chunk/cite/get_paper_figures di; high: Weakest claim 2 not hardened: d-11 blocked, no verbatim citations for large lepton asymmetry L~1e-2, secret interaction G_X, or low T_RH thresholds and their independent BBN/CMB free-streaming constraints; contrasting suppressed-thermalization emulator grid (Delta N_eff=0.2-0.3) not run.

- `c-09` (wave 2, report): **ok**

B.5 Open directions

1. `d-06` Quantify non-standard loopholes required to reconcile eV sterile with N_eff=2.99+/-0.17 and their independent constraints — Standard thermalization is excluded at 6.2 sigma but loophole viability remains unproven and is the only path to reconcile SBL with cosmology

2. `d-04` Complete literature survey 2024-2026 SBL status with 5 verbatim citations including arXiv:2504.16590 and 1-2 extracted figures — SBL appearance vs disappearance status currently rests on uncitable snippets and one gallium citation

3. `d-03` Thermalization threshold, exclusion plane overlay, and viability assessment with non-standard loopholes and eV vs keV contrast — Numerical factors 7-313x above allowed mixing exist but visual overlay was blocked by tool limits

4. `d-02` Literature survey of SBL anomaly status 2024-2026: MicroBooNE, STEREO, PROSPECT, BEST/gallium, reactor flux re-evaluation, global 3+1 fits — Gallium persists at >5 sigma but tension with nulls determines whether sterile explanation remains viable